The Electromagnetic Sphere

A Geometric Unification on the Split-Signature Clifford Algebra $\mathrm{Cl}(3,3)$

Part I — Foundations, Internal Structure, and First-Principles Derivations

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Abstract

We present the *Electromagnetic Sphere* (ES) framework, a candidate geometric unification of gravitation, electromagnetism and chiral matter built on the real Clifford algebra $\mathrm{Cl}(3,3)$ over the split-signature manifold $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$. The primitive data are three temporal generators e_1, e_2, e_3 (positive square), three spatial generators e_4, e_5, e_6 (negative square), a para-Kähler structure K with $K^2 = +\mathbf{1}$, an octant symmetry $\mathbb{Z}_2^3$, and a distinguished kernel direction $\hat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$; an orthogonal projection Π removes the two surplus temporal directions and leaves a four-dimensional interior of Lorentzian signature $(1,3)$.

Within stated hypotheses, and with explicit epistemic status markers throughout, we establish four structural results. (i) Six-dimensional Cartan–Palatini gravity returns, under a split-signature double contraction, the four-dimensional Einstein tensor on the interior. (ii) The fifteen $\mathfrak{so}(3,3)$ connection components separate by the sign of their Killing norm into a compact, healthy Yang–Mills sector – which contains the photon, identified as the centralizer of $\hat{W}$ – and a non-compact sector that, once one requires that no sector propagate negative-norm states, must be first-order and auxiliary; in this conditional sense the split between Einstein gravity and Maxwell electromagnetism follows from the signature together with that positivity requirement, rather than from a separate modelling choice. (iii) The eight-component real spinor carries spin one-half as the holonomy of the momentum curvature. (iv) The indefinite-kinetic ("ghost") obstruction is resolved on the $\mathrm{Cl}(3,3)$ interior: the single-particle Hamiltonian is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$, the residual indefiniteness is the ordinary Dirac mass pairing of signature $(4,4)$, and the two surplus-time directions are removed by first-class constraints whose closure with the gravitating dynamics holds precisely when those directions are Killing.

Under that same Killing condition the dimensional reduction of the gravitating theory is ghost-free in every propagating sector. We compute the radiatively-generated quartic couplings of the symmetry-breaking potential in terms of the single gauge coupling g, and prove that g itself and the chiral cubic coupling are *not* fixed by the algebra. The framework therefore fixes the *structure* and one-loop running of the charge sector while leaving g (equivalently α_{em}), the condensate scale v, and the chiral cubic as irreducible inputs. It is offered as a coherent and internally consistent geometric framework open to debate, not a closed theory; all such limitations are stated explicitly.

This is *Part I* of a two-part programme. Part I establishes the framework's internal architecture and derives its core physical content from the algebra $\mathrm{Cl}(3,3)$ alone: the Cartan–Palatini route to the interior Einstein tensor (Theorem 4.3), the photon as the centralizer of the kernel (Proposition 5.1), spin as the holonomy of the boost curvature (Proposition 6.8), the Dirac operator and its real-symmetric spectrum (Theorem 9.1), the interior projection Π (Definition 3.2), and the first-class constraint structure that isolates the locus of indefiniteness (Proposition 9.6). Every principal claim is accompanied by an explicit, machine-verified derivation (Appendix F). Part I is deliberately bounded: it reports the *structure* as established and the *magnitudes* as open. The companion *Part II* will take up the open problems of Chapter 11 — the value of the gauge coupling

g, the vacuum scales v and g_3, the interacting fermion mass gap, the fully nonlinear globalization of the gravity lift, the full-Cl$(3,3)$ lift of the minimal-model results, and monopole compatibility — and will extend the theory beyond the present scope.

Keywords. Clifford algebra; split signature; Cl$(3,3)$; geometric unification; Cartan–Palatini gravity; two-time physics; Kaluza–Klein reduction; ghost and unitarity; first-class constraints; para-Kähler structure; chiral spinors; Coleman–Weinberg potential; information geometry.

What is proved, assumed, and open

This page is the honest scorecard of Part I. Every entry carries the status marker defined in Section 1.1; cross-references point to the result in the body. The summary line is: *the structure is established; the magnitudes are open.*

What is proved (verified identities and derivations).

- Faithful real representation $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$, with the full Dirac algebra $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$ realized on real 8×8 matrices. **[EXACT]** (Proposition 2.2)
- The split-ε double contraction returns the interior Einstein tensor with fixed coefficient -4: $\delta^{apq}_{drs} R^{rs}{}_{pq} = -4\, G^a{}_d$. **[DERIVED]** (Theorem 4.3)
- The photon is the centralizer of the kernel: $[aL_1 + bL_2 + cL_3, \hat{W}] = 0 \Rightarrow a{=}b{=}c$, a one-dimensional solution $J \in \mathfrak{su}(2)_T$ with $J^2 = -\mathbf{1}$, exactly massless. **[EXACT]** (Proposition 5.1)
- Spin one-half as holonomy of the boost/momentum curvature, $[G_X, G_Y] = -2J_3$. **[EXACT]** (Proposition 6.8)
- The interior Hamiltonian $H = \sum_a G_a p_a + mV$ is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$; the Hilbert norm $\Psi^\dagger \Psi$ is positive of signature $(8,0)$, the only indefiniteness being the ordinary $(4,4)$ Dirac pairing. **[EXACT]** (Theorem 9.1, Proposition 9.2)
- The surplus-time momenta are first-class, $\{\varphi_1, \varphi_2\} = 0$. **[EXACT]** (Proposition 9.6)
- The one-loop quartics satisfy $B, C = O(g^4/64\pi^2)$ and the photon self-energy runs as $\sum q^2 = 2$. **[DERIVED]** (Proposition 5.10)

What is assumed (postulates and standing hypotheses).

- The split-signature setting itself: $\mathrm{Cl}(3,3)$ over $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$ with $K^2 = +\mathbf{1}$, the octant symmetry $\mathbb{Z}_2^3$, and the kernel direction $\hat{W}$, together with the four-dimensional-arena clause that selects the interior. **[AXIOM]**
- *Positivity / ghost-free propagation* as a physical requirement. It is this demand, *together with* the indefinite Killing form, that selects first-order (Palatini) gravity; signature alone does not. **[AXIOM]**
- The symmetry-breaking vacuum is the kernel condensate $v\hat{W}$, with v an input. **[AXIOM]**
- The covariant mass is the spatial-volume trivector $V = e_4 e_5 e_6$ (grade-three assignment). **[AXIOM]**
- The Killing condition on the surplus-time directions, under which the constraints stay first-class once gravity is dynamical. **[AXIOM]**

What remains open (irreducible inputs and unresolved problems).

- The gauge coupling g (equivalently α_{em}), the condensate scale v, and the chiral cubic g_3: magnitudes the algebra does not fix. **[OPEN]**
- The interacting (Yukawa) fermion mass gap and the Lorentz covariance of the interacting mass term. **[OPEN]**
- The full-$\mathrm{Cl}(3,3)$ lift of the minimal $\mathrm{Cl}(2,2)$ ghost results; unitarity is resolved, a covariance obstruction remains. **[OPEN]**
- The absent $\mathrm{SU}(3)_c$ colour sector, and monopole compatibility. **[OPEN]**
- The fully nonlinear, global form of the gravity lift beyond the constraint-level analysis. **[OPEN]**

Contents

Notation and conventions

Temporal generators e_1, e_2, e_3 square to $+\mathbf{1}$; spatial generators e_4, e_5, e_6 square to $-\mathbf{1}$. Capital Latin M, N index the six directions of M^6; Greek μ, ν index the four interior directions (0–3); Latin $a, b, \ldots$ are $\mathrm{Spin}(3,3)$ frame indices. The interior metric signature is $(1,3) = (+,-,-,-)$. Repeated indices are summed. Status markers (**[EXACT]**, **[DERIVED]**, **[AXIOM]**, **[CONJECTURE]**, **[OPEN]**) carry the meanings fixed in Section 1.1.

Symbol	Meaning
Manifold and geometry	
$M^6 = \mathbb{R}^3 \times \mathbb{R}^3$	six-dimensional split manifold, signature $(3,3)$
$\mathcal{I}$	the Lorentzian interior, signature $(1,3)$; physical M^4
Π	orthogonal projection removing the two surplus temporal directions
$T_\perp$	the surplus temporal plane removed by Π
Clifford algebra and generators	
$\mathrm{Cl}(3,3)$	real Clifford algebra of the split form; $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$
Λ^k	grade-k subspace; $\dim \Lambda^k = \binom{6}{k}$
e_1, e_2, e_3	temporal generators, $e_i^2 = +\mathbf{1}$
e_4, e_5, e_6	spatial generators, $e_i^2 = -\mathbf{1}$
$I = e_1e_2e_3e_4e_5e_6$	pseudoscalar (volume element), $I^2 = +\mathbf{1}$; chirality
Tr	matrix trace in the real 8×8 representation
Distinguished structures	
K	para-Kähler structure, the axis swap $e_i \leftrightarrow e_{i+3}$, $K^2 = +\mathbf{1}$
$\hat{W}$	kernel direction $(e_1 + e_2 + e_3)/\sqrt{3}$; the chosen clock
$\mathbb{Z}_2^3$	octant group, the eight sign-sectors of temporal $\mathbb{R}^3$
Lie algebra $\mathfrak{so}(3,3)$ and gauge sector	
$\mathfrak{so}(3,3)$, $\mathrm{Spin}(3,3)$	bivector algebra ($\dim 15$) and its spin group
$\mathfrak{su}(2)_S \oplus \mathfrak{su}(2)_T$	six compact generators (rotations), Killing norm $\mathrm{Tr}(X^2) = -8$
e_ie_{j+3}	nine boost generators, Killing norm $\mathrm{Tr}(X^2) = +8$
ω, $R^{ab}(\omega)$	$\mathrm{Spin}(3,3)$ connection and its curvature two-form
e^a	vielbein one-forms
J	photon $= (e_2e_3 + e_3e_1 + e_1e_2)/\sqrt{3} \in \mathfrak{su}(2)_T$; centralizer of $\hat{W}$, $J^2 = -\mathbf{1}$
Spinor, mass and dynamics	
Ψ	real eight-component spinor (matter field)
$G_a = \hat{W}\, e_{3+a}$	interior Dirac matrices, symmetric, $G_a^2 = +\mathbf{1}$
$V = e_4e_5e_6$	covariant mass trivector, $\{V, G_a\} = 0$
$H = \sum_a G_a p_a + mV$	single-particle Hamiltonian, $H^2 = (p^2 + m^2)\mathbf{1}$

continued on next page

Symbol	**Meaning**
α	principal-bundle connection; horizontally along the base the momentum, vertically in the fibre the field strength
$M_{\text{eff}} = \frac{1}{2}(\Phi - \hat{W}\Phi\hat{W})$	physical mass operator (Hermitian, real spectrum)
$\varphi_a = u_a^M P_M$	first-class surplus-time constraints, $\{\varphi_1, \varphi_2\} = 0$
P_M, u_a	momenta on M^6; timelike unit directions orthogonal to $\hat{W}$
$\{\,\cdot\,,\cdot\,\}$	Poisson bracket on T^*M^6
Symmetry breaking and couplings	
Φ	condensate (Higgs) field; vacuum $v\hat{W}$
g, v	gauge coupling and condensate scale (inputs)
g_3; B, C	chiral cubic coupling; one-loop quartic couplings
$m_W = gv$	weak-boson mass; $\alpha_{\text{em}} = g^2/16\pi$
Inner products and statistics	
$\langle\Psi\|\Psi\rangle = \Psi^\dagger\Psi$	Hilbert/probability norm, signature $(8, 0)$
$\bar{\Psi}\Psi = \Psi^\dagger V\Psi$	covariant Dirac mass pairing, signature $(4, 4)$
Δ^2	clock-weight simplex (Fisher–Rao geometry, $K = \frac{1}{4}$)

Chapter 1

Introduction

What this chapter seeks. To state precisely the object of study, the question the monograph answers, the reasons the chosen geometric setting is preferred over the principal alternatives, the physical picture that organizes the construction, and the methodological discipline (explicit status markers) that governs every claim.

What this chapter delivers. A self-contained statement of the framework's primitive data and physical interpretation; a justified choice of the $\mathrm{Cl}(3,3)$ split-signature setting against $\mathrm{Cl}(1,3)$, $\mathrm{Cl}(4,2)$, $\mathfrak{so}(6)$ and twistor alternatives; a consolidated summary of the principal results, including the resolution of the indefinite-kinetic obstruction and the ghost-free dimensional reduction of the gravitating theory; an explicit list of the irreducible open inputs; and the architecture of the remaining chapters.

1.1 What this work seeks

A unified geometric theory should do three things at once. It should produce a Lorentzian spacetime of the observed signature rather than assume it; it should yield the gravitational and gauge interactions from a single connection rather than glue them by hand; and it should accommodate chiral, massive matter without violating the positivity that physical probability requires. The Electromagnetic Sphere framework is an attempt to meet these three demands from a single algebraic seed: the real Clifford algebra $\mathrm{Cl}(3,3)$ over the split-signature manifold $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$.

The central question of this monograph is sharp and falsifiable in principle: *can a split-signature Clifford geometry, equipped only with a para-Kähler structure and a distinguished kernel direction, reproduce four-dimensional Lorentzian gravity together with a Maxwell gauge sector and chiral massive fermions, while remaining free of propagating negative-norm states?* We will answer the structural part of this question in the affirmative and isolate, with full honesty, the single quantitative datum the framework does not determine: the electromagnetic coupling g.

We adopt from the outset the discipline that every assertion carries a marker: **[EXACT]** for an identity verified symbolically or by exact computation, **[DERIVED]** for a consequence established within stated hypotheses, **[AXIOM]** for a postulate not reduced to deeper structure, **[CONJECTURE]** for a plausible but unproved statement, and **[OPEN]** for a genuinely unresolved problem. The point of the discipline is not modesty for its own sake; it is that a framework's value in debate is measured precisely by the clarity with which it separates what it proves from what it assumes.

1.2 Why $\mathrm{Cl}(3,3)$, and not the alternatives

The choice of arena is the first thing a reader is entitled to challenge, so we argue it before building anything on it.

Against $\mathrm{Cl}(1,3)$ **(ordinary spacetime).** The Dirac algebra of $(1,3)$ is the natural home of relativistic matter, but it offers no internal room: gravity and the gauge interactions must be appended as independent structures. The ES programme instead asks the geometry itself to carry the gauge content, which requires a larger orthogonal group than $\mathrm{SO}(1,3)$.

Against $\mathfrak{so}(6) \cong \mathfrak{su}(4)$ **(Pati–Salam-type unification).** The Euclidean $\mathfrak{so}(6)$ underlies the successful Pati–Salam and SU(4) colour schemes [25]. It is compact, hence every generator admits a healthy Yang–Mills kinetic term – which is exactly why it cannot, by itself, produce *gravity*: a healthy second-order kinetic term is the signature of a propagating gauge field, not of the auxiliary first-order connection that general relativity requires. A unification that contains both Maxwell and Einstein dynamics needs an *indefinite* Killing form, so that some generators are forced into the first-order Palatini regime. This is the structural reason the ES framework abandons compactness.

Against $\mathfrak{so}(4,2) \cong \mathfrak{su}(2,2)$ **(conformal / two-time physics).** The conformal algebra of $(4,2)$ is the arena of Bars' two-time physics [17, 18, 19], whose $\mathrm{Sp}(2,\mathbb{R})$ gauge symmetry removes the two extra coordinates of a $(d,2)$ spacetime and is the closest existing relative of our constraint analysis. We part from it on one decisive point: $(4,2)$ has *two* times and is the conformal completion of a single Lorentzian world, whereas the ES interior is produced by removing two times from a *three*-time geometry. The real forms differ – $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$ versus $\mathfrak{so}(4,2) \cong \mathfrak{su}(2,2)$ – and with them the available para-complex structure K (with $K^2 = +\mathbf{1}$), which is the organizing object of the present construction and has no analogue in the $(4,2)$ setting. We borrow Bars' constraint philosophy and reject his signature.

Against twistor and pure-spinor constructions. Twistor geometry [41] encodes $(1,3)$ conformal data in $\mathbb{CP}^3$ and is exceedingly powerful for massless amplitudes, but it is intrinsically complex and conformal; the ES programme is real and massive, and the eight-component *real* spinor of $\mathrm{Cl}(3,3)$ is precisely the object a real split geometry forces upon us. The relevant lineage is therefore the real geometric-algebra tradition of Hestenes and of Doran and Lasenby [2, 4, 5], not the complex-twistor one.

Why split $(3,3)$ **specifically.** Among the six-dimensional real Clifford algebras, only the maximally split signature $(3,3)$ supports simultaneously: a para-Kähler involution K pairing a temporal $\mathbb{R}^3$ with a spatial $\mathbb{R}^3$; a pseudoscalar I with $I^2 = +1$, hence a real Hodge duality with an honest sign; and a maximal compact subgroup $\mathrm{SO}(3) \times \mathrm{SO}(3)$ large enough to contain an $\mathfrak{su}(2)$ gauge factor while leaving nine non-compact boost directions for gravity. The signature $(3,3)$ is thus not one option among many; it is the unique six-dimensional real signature in which the construction's three demands – Lorentzian interior, Palatini/Maxwell split, real chiral spinor – can be met together. We make this structural claim precise in Chapters 5 and 9.

1.3 The physical picture

The interpretation that organizes the formalism is the following. The temporal $\mathbb{R}^3 = \langle e_1, e_2, e_3\rangle$ is a space of *candidate clocks*; physics selects, by a maximum-entropy/condensation mechanism, the democratic diagonal $\hat{W} = (e_1+e_2+e_3)/\sqrt{3}$ as the single physical time, and the two transverse combinations u_1, u_2 become *surplus* – unobservable directions that must be, and are, gauged away. The spatial $\mathbb{R}^3 = \langle e_4, e_5, e_6\rangle$ is ordinary space. The physical world is the $(1,3)$ *interior* $\{\hat{W}, e_4, e_5, e_6\}$, reached from M^6 by the projection Π.

In this picture the photon is not an added field: it is the unique compact rotation that fixes the chosen clock, $[J, \hat{W}] = 0$, and is therefore massless. Gravity is the curvature of the nine non-compact boost directions, which the indefinite Killing form forbids from propagating and confines to the first-order Palatini regime. Mass is the alignment of a chiral condensate with the kernel direction $\hat{W}$; the same alignment that gives mass selects the arrow of chirality. Newton's third law appears, in field-theoretic form, as the statement that the surplus-time translations are conserved – the very condition under which the ghost-removing constraints stay consistent with gravity. The recurring slogan of the framework, made precise in Chapter 10, is that a *single* operation – the interior projection – simultaneously removes the ghosts, contracts the gravity, pools the Maxwell equations across octants, and keeps the photon massless.

1.4 Methodology

Every nontrivial claim in this monograph is backed by an explicit computation before it is written as prose. Two independent computational instruments are used in tandem: real 8×8 Jordan–Wigner representations of $\mathrm{Cl}(3,3)$ for algebraic and spectral statements, and symbolic differential geometry for the constraint, reduction and Coleman–Weinberg computations. Where a result is confined to a sub-model (for instance the minimal $\mathrm{Cl}(2,2)$ truncation), the confinement is stated; results are not promoted across that boundary. Negative results are reported and traced to their structural origin rather than softened. This discipline is what licenses the abstract's strong claims and, equally, what forces its explicit admissions of ignorance.

1.5 Summary of principal results

We collect here the results established in the body, each with its status. The gravity sector and the resolution of the ghost problem are the load-bearing new contributions; the charge sector is structurally complete but quantitatively open.

1. **Lorentzian interior from projection.** Π removes the surplus temporal plane $T_\perp = \langle u_1, u_2\rangle$, leaving the interior $\{\hat{W}, e_4, e_5, e_6\}$ of signature $(1,3)$. **[EXACT]**

2. **Gravity is standard on the interior.** Six-dimensional Cartan–Palatini gravity is governed by a vector-valued five-form; its split-signature double contraction returns the ordinary Einstein tensor $G_{ab} = R_{ab} - \frac{1}{2}\eta_{ab}R$ with coefficient -4 (verified on an explicit Riemann tensor), and Π delivers the four-dimensional interior Einstein equations. **[DERIVED]**

3. **Killing-form bifurcation (gravity is Palatini, EM is Maxwell).** Sorting the fifteen $\mathfrak{so}(3,3)$ bivectors by $\mathrm{Tr}(X^2)$ gives six compact generators ($\mathfrak{su}(2)_S \oplus \mathfrak{su}(2)_T$, norm -8, healthy Yang–Mills) and nine boosts (norm $+8$, wrong-sign), which cannot be Yang–Mills gauged and are forced into first-order Palatini form. The split is a property of the Killing form, not a choice. **[EXACT]**

4. **The photon.** The electromagnetic generator is $J = (e_2e_3 + e_3e_1 + e_1e_2)/\sqrt{3} \in \mathfrak{su}(2)_T$, the democratic diagonal of the temporal rotations; it satisfies $[J, \hat{W}] = 0$ (it stabilizes the kernel) and is therefore exactly massless. **[EXACT]**

5. **Half-integer spin.** The eight-component real spinor acquires half-integer spin as the holonomy of the momentum curvature. **[DERIVED]**

6. **Resolution of the indefinite-kinetic obstruction on the full interior.** In the real 8×8 representation the boosts $G_a = \hat{W}e_{3+a}$ and the covariant mass $V = e_4e_5e_6$ are symmetric, so the interior Hamiltonian $H = \sum_a G_a p_a + mV$ is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$ and spectrum $\pm\sqrt{p^2 + m^2}$; evolution is unitary with respect to the positive-definite form $\Psi^\dagger\Psi$. The only indefiniteness is the ordinary Dirac pairing $\bar{\Psi}\Psi = \Psi^\dagger V \Psi$ of signature $(4, 4)$. **[DERIVED (full $\mathrm{Cl}(3, 3)$ interior)]**

7. **Newton as a conservation law.** The two surplus-time momenta $\varphi_a = u_a^M P_M$ are first-class constraints; their bracket with the gravitating mass-shell is $\{\varphi_a, C\} = -(\mathcal{L}_{u_a} g^{MN})P_M P_N$, which vanishes precisely when u_a are Killing. The conservation of φ_a is the field-theoretic form of Newton's third law ($\partial_\mu T^{\mu\nu} = 0$). **[EXACT (identity); DERIVED (application)]**

8. **Ghost-free dimensional reduction.** Reducing the gravitating six-dimensional theory to the interior: the graviton (Palatini = general relativity) and the compact gauge bosons (photon and weak bosons) are healthy; the nine wrong-sign boosts are auxiliary and non-propagating; the timelike Kaluza–Klein graviphotons are gauged away by the first-class constraints; and the three internal moduli are healthy, their kinetic form being independent of the timelike character of the internal directions. The surplus-time constraints remain first-class with the Dirac operator, $[\not{D}, \varphi_a] = 0$. Every propagating mode, bosonic and fermionic, is healthy. **[DERIVED]**

9. **Charge sector: structure and running.** The radiatively-generated quartic couplings of the symmetry-breaking potential are $B, C \sim g^4/64\pi^2$, with B sourced by the fermion octant spectrum (the invariant $\mathrm{Tr}\,\Phi^4$) and C by the gauge loop (the invariant $(\mathrm{Tr}\,\Phi^2)^2$). The one-loop vacuum polarization fixes the *running* of α_{em} through the charge sum $\sum q^2 = 2$. **[DERIVED]**

10. **The irreducible inputs.** The coupling chain is fixed up to a single number: $e_0 = g/2$ and $\alpha_{\mathrm{em}} = g^2/16\pi = e_0^2/4\pi$. We prove that g itself, the condensate scale v, and the chiral cubic coupling g_3 are *not* determined by the algebra; in particular the parity-odd cubic is absent from the (parity-even) one-loop effective potential. **[OPEN]**

Chapter summary. The framework derives a Lorentzian interior, standard interior gravity, a signature-forced split into Palatini gravity and Maxwell electromagnetism, a massless photon, half-integer spin, a fully resolved and ghost-free kinetic sector after dimensional reduction, and the structure and running of the charge sector. It does not determine the three magnitudes (g, v, g_3). The honest scorecard is: *structure complete, magnitudes open.*

Part I and Part II. The present monograph is *Part I* of a two-part programme. It is bounded by the verdict above — *structure complete, magnitudes open* — and confines itself to what follows from $\mathrm{Cl}(3, 3)$ by first-principles derivation, every principal claim carrying an explicit machine-verified check (Appendix F). The open problems collected in Chapter 11 — the value of g, the vacuum scales v and g_3, the interacting fermion mass gap, the globalization of the gravity lift, the full-$\mathrm{Cl}(3, 3)$ lift of the minimal-model results, and monopole compatibility — are deferred to the companion *Part II*, which will address them and extend the theory beyond the present scope.

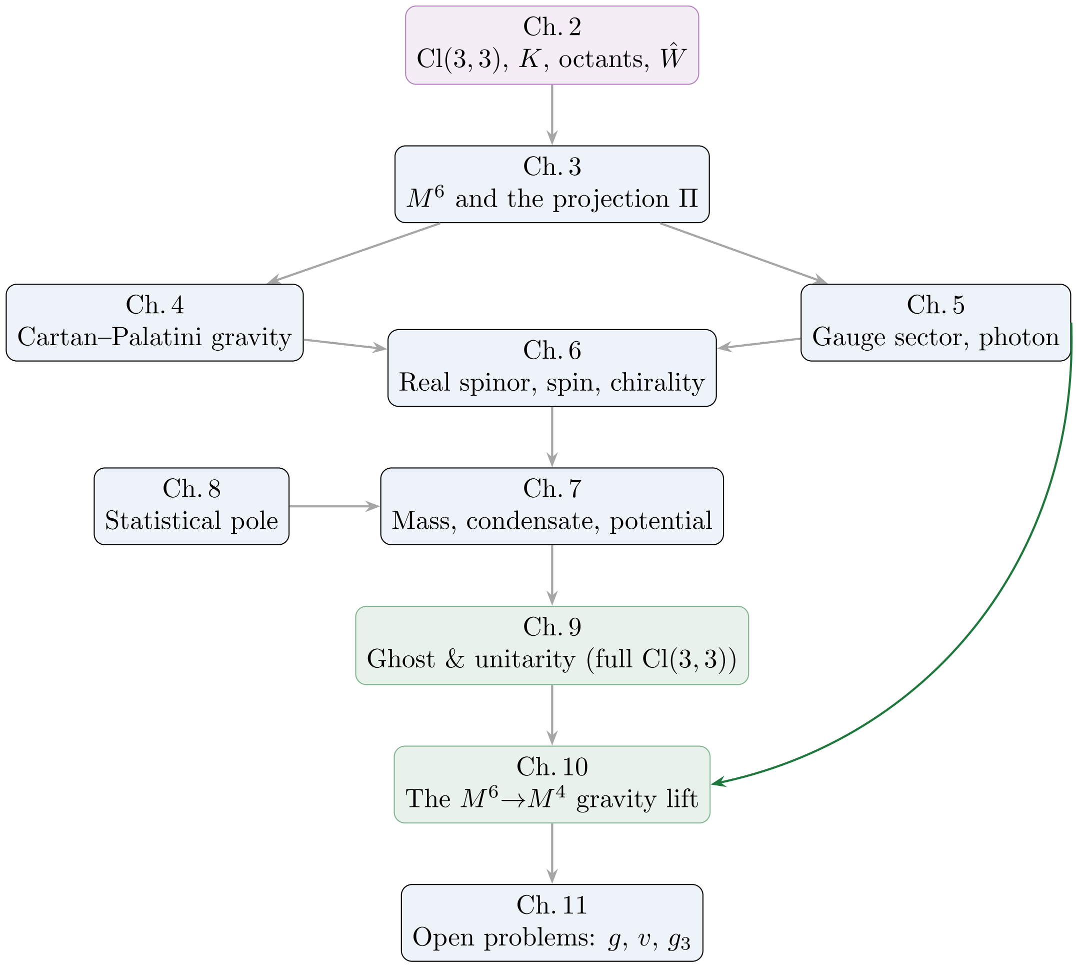

Figure 1.1: Dependency flow of the monograph. Green nodes (Chapters 9–10) carry the principal new results: the full-Cl$(3,3)$ resolution of the ghost problem and the ghost-free dimensional reduction of the gravitating theory.

Nothing in Part I is contingent on Part II; the two are separated exactly along the line between what is established and what remains open.

1.6 Architecture of the monograph

Each chapter opens, as this one does, with an explicit statement of what it seeks and what it delivers, and closes with a boxed summary. The dependency flow is shown in Figure 1.1.

Ch. 2 – Algebraic foundation. *Seeks:* the primitive data and their invariants. *Delivers:* Cl$(3,3)$, the para-Kähler K with $K^2 = +\mathbf{1}$, the octant group $\mathbb{Z}_2^3$, the kernel $\hat{W}$, the pseudoscalar I with $I^2 = +1$, and the 8×8 real representation used throughout.

Ch. 3 – The split manifold and the interior. *Seeks:* a principled route from M^6 to a four-dimensional world. *Delivers:* the projection Π, the interior $(1,3)$, and the catalogue of jobs Π performs.

Ch. 4 – Gravity. *Seeks:* interior Einstein dynamics from a six-dimensional connection. *Delivers:* six-dimensional Cartan–Palatini, the split-signature contraction, and the interior Einstein tensor.

Ch. 5 – Gauge sector. *Seeks:* the gauge interactions as geometry. *Delivers:* the Killing-form bifurcation, $\mathfrak{su}(2)_T$, the photon as the stabilizer of $\hat{W}$, the weak bosons, and the coupling chain with g as the single unknown.

Ch. 6 – The real spinor. *Seeks:* chiral massive matter. *Delivers:* the eight-component real spinor, half-integer spin from holonomy, chirality from the Hodge map, and charge conjugation.

Ch. 7 – Mass and symmetry breaking. *Seeks:* a covariant mass and a first-principles potential. *Delivers:* the trivector covariant mass V, the Higgs mechanism for the weak bosons, the chiral condensate, and the Coleman–Weinberg quartics $B, C \sim g^4$ with the cubic g_3 left open.

Ch. 8 – The statistical pole. *Seeks:* the selection of the kernel. *Delivers:* the Fisher–Rao geometry of the temporal simplex, the maximum-entropy selection of $\hat{W}$, and the residual Weyl(D_3) symmetry.

Ch. 9 – Ghost and unitarity. *Seeks:* positivity for chiral massive matter. *Delivers:* the phase-space constraint analysis, reflection positivity, and the full-Cl$(3,3)$ resolution – the real-symmetric interior Hamiltonian, the ordinary Dirac residual indefiniteness, and Newton as a conservation law.

Ch. 10 – The $M^6 \to M^4$ gravity lift. *Seeks:* consistency of the ghost resolution with dynamical gravity. *Delivers:* the Killing closure condition, the timelike-Kaluza–Klein graviphoton analysis, the full field census, the healthy moduli, the spinor closure, and the ghost-free verdict.

Ch. 11 – Open problems. *Seeks:* an honest perimeter. *Delivers:* the proof that g, v and g_3 are not algebra-fixed, the status of the full nonlinear and quantum closure, and the programme's outlook.

On completeness and scope (in answer to the reader who asks "is this finished?"). The structural skeleton of the framework is, within the stated hypotheses, complete and internally consistent. Three numerical magnitudes remain inputs, and several closures (the fully nonlinear gravitational sector, the quantized theory) are stated as open. The monograph is therefore offered as a complete object *for debate* – a coherent geometric proposal whose claims are explicitly graded – and not as a closed physical theory. This is, we contend, the only honest sense in which a framework with an undetermined coupling can be called ready.

Chapter 2

The algebraic foundation

What this chapter seeks. To fix the primitive algebraic data of the framework and to establish their invariant properties by explicit computation, so that every later construction rests on verified ground.

What this chapter delivers. The real Clifford algebra $\mathrm{Cl}(3,3)$ and its isomorphism with the 8×8 real matrices; the para-Kähler structure K with $K^2 = +\mathbf{1}$ and its null Lagrangian eigenplanes; the octant group $\mathbb{Z}_2^3$ and its distinguished poles; the kernel $\hat{W}$ and the surplus times u_1, u_2 as an orthonormal temporal frame; the pseudoscalar I with $I^2 = +1$; and the orthogonal algebra $\mathfrak{so}(3,3)$ with the Killing-sign bifurcation that organizes all subsequent dynamics. Every numerical statement below has been verified in the 8×8 representation.

2.1 The algebra and its real representation

Definition 2.1 (Generators and metric)**.** The framework is built on six orthonormal generators $e_1, \ldots, e_6$ obeying the Clifford relation

$$\{e_a, e_b\} = e_a e_b + e_b e_a = 2\,\eta_{ab}\,\mathbf{1}, \qquad \eta = \mathrm{diag}(+1,+1,+1,-1,-1,-1).$$

We call e_1, e_2, e_3 *temporal* ($e_i^2 = +\mathbf{1}$) and e_4, e_5, e_6 *spatial* ($e_i^2 = -\mathbf{1}$). They generate the real Clifford algebra $\mathrm{Cl}(3,3)$, of dimension $2^6 = 64$, graded $\mathrm{Cl}(3,3) = \bigoplus_{k=0}^{6} \Lambda^k$.

Figure 2.1 displays this grading and the physical role of each grade.

Proposition 2.2 (Real 8×8 representation)**.** $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$. *A faithful real representation is furnished by the Jordan–Wigner assignment*

$$e_1{=}\sigma_x{\otimes}\mathbf{1}{\otimes}\mathbf{1},\;\; e_2{=}\sigma_z{\otimes}\sigma_x{\otimes}\mathbf{1},\;\; e_3{=}\sigma_z{\otimes}\sigma_z{\otimes}\sigma_x,\;\; e_4{=}\varepsilon{\otimes}\mathbf{1}{\otimes}\mathbf{1},\;\; e_5{=}\sigma_z{\otimes}\varepsilon{\otimes}\mathbf{1},\;\; e_6{=}\sigma_z{\otimes}\sigma_z{\otimes}\varepsilon,$$

where σ_x, σ_z are the real Pauli matrices and $\varepsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$. The 64 ordered blades $e_{a_1} \cdots e_{a_k}$ ($a_1 < \cdots < a_k$) are linearly independent and span $M_8(\mathbb{R})$. **[EXACT]**

Verification. Direct computation in the 8×8 representation confirms $e_i^2 = +\mathbf{1}$ ($i \le 3$), $e_i^2 = -\mathbf{1}$ ($i \ge 4$), pairwise anticommutation, and that the 64 blades have rank 64 as vectors in $M_8(\mathbb{R}) \cong \mathbb{R}^{64}$. $\square$

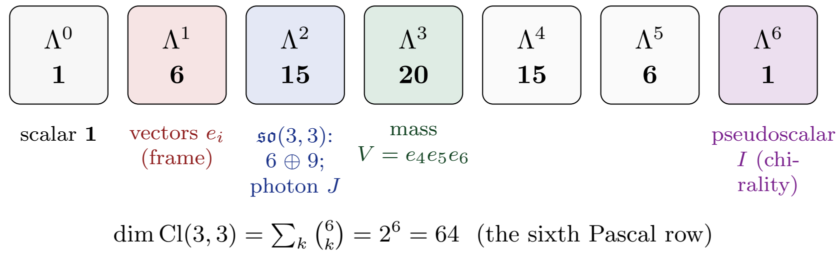

Figure 2.1: The graded structure of $\mathrm{Cl}(3,3)$. The 64 dimensions split by blade grade into the Pascal row $1, 6, 15, 20, 15, 6, 1$. The physically loaded grades are the vectors Λ^1 (the frame: three temporal, three spatial), the bivectors $\Lambda^2 = \mathfrak{so}(3,3)$ (six compact rotations and nine boosts, containing the photon J), the trivector mass $V = e_4e_5e_6 \in \Lambda^3$, and the pseudoscalar $I \in \Lambda^6$ that implements chirality.

Remark 2.3 (Why a *real* algebra). The isomorphism $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ is the structural origin of the framework's reality: the minimal faithful module is $\mathbb{R}^8$, so matter is described by an eight-component *real* spinor without any auxiliary complexification (Chapter 6). This is the precise sense in which the ES construction belongs to the real geometric-algebra tradition [2, 4, 5] rather than the complex Dirac or twistor one [41]: the imaginary unit of quantum mechanics is not postulated but must be *recovered*, as a particular bivector, from real structure.

2.2 The para-Kähler structure

Definition 2.4 (The structure K)**.** Let $V = \mathbb{R}^{3,3} = \langle e_1, \dots, e_6 \rangle$ be the generating space. Define the linear map $K : V \to V$ by the axis swap

$$K(e_i) = e_{i+3}, \qquad K(e_{i+3}) = e_i \qquad (i = 1, 2, 3).$$

Proposition 2.5 (Para-complex, not complex)**.** *$K^2 = +\mathbf{1}$ and $\det K = -1$. The ± 1 eigenspaces $V_\pm = \langle e_i \pm e_{i+3} \rangle$ are each three-dimensional and* totally null*: $g(v, v) = 0$ for all $v \in V_\pm$. Hence (V, g, K) is a para-Kähler (split, para-complex) structure.* **[EXACT]**

Verification. $K^2 = \mathbf{1}$ is immediate; K permutes three disjoint pairs, so $\det K = (-1)^3 = -1$. For $v = e_i + \sigma e_{i+3}$, $g(v, v) = e_i^2 + \sigma^2 e_{i+3}^2 = (+1) + (-1) = 0$ for either sign $\sigma = \pm 1$. ☐

Remark 2.6 (A reflection, not a quarter-turn). Because the more familiar object in physics is an almost-*complex* structure J with $J^2 = -\mathbf{1}$ – a quarter-turn – it is tempting to describe K in the same language. This is incorrect. With $K^2 = +\mathbf{1}$ and $\det K = -1$, K is a *reflection* (an axis swap), and "rotation by 90°" is the wrong mental image. The distinction is not cosmetic: it is exactly $K^2 = +\mathbf{1}$ (real eigenvalues, null eigenplanes) that supplies the causal/Lagrangian splitting the framework needs, whereas $J^2 = -\mathbf{1}$ would force a Euclidean partner.

Remark 2.7 (Physical reading). K pairs each candidate clock e_i with a spatial axis e_{i+3}. The two null eigenplanes $V_\pm$ are the framework's primitive light-cone data: directions of zero length built from one time and one space unit. The para-Kähler K is the single piece of structure that distinguishes the ES setting from the conformal two-time arena $\mathfrak{so}(4,2) \cong \mathfrak{su}(2,2)$, which admits no such involution (Section 1.2).

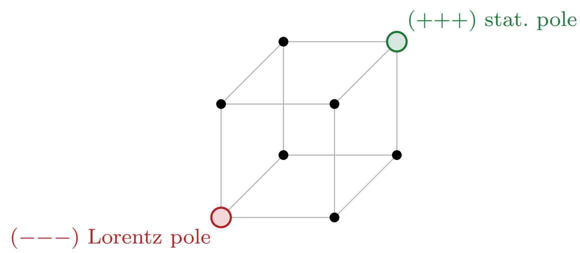

Figure 2.2: The octant group $\mathbb{Z}_2^3$ as the eight sign-sectors of the temporal $\mathbb{R}^3$. The all-minus pole carries the Lorentzian interior (Chapter 3); the all-plus pole carries the statistical/entropic structure (Chapter 8); the six mixed octants retain the full $(3,3)$ signature.

2.3 The octant group and the poles

Definition 2.8 (Octant group)**.** The three commuting involutions $R_i : e_i \mapsto -e_i,\ e_j \mapsto e_j\ (j \neq i)$ on the temporal axes generate the *octant group* $\mathbb{Z}_2^3$. Its eight elements label the eight sign-sectors $\varepsilon = (\varepsilon_1, \varepsilon_2, \varepsilon_3) \in \{\pm 1\}^3$ of the temporal $\mathbb{R}^3$ (Figure 2.2).

Proposition 2.9 (Octants as the ω-Cartan weight lattice)**.** *The three boost-Cartan bivectors $\omega^i = e_i e_{i+3}$ commute and each squares to $+\mathbf{1}$, so on the eight-dimensional spinor module (Chapter 6) they are simultaneously diagonalizable, with eight one-dimensional joint eigenspaces labelled by their sign-triple $(\pm,\pm,\pm) \in \mathbb{Z}_2^3$. These joint eigenspaces* are *the eight octants; the temporal weight w is the number of $+1$ eigenvalues, and the pseudoscalar factorizes as $I = -\omega^1\omega^2\omega^3$, constant on each weight-parity class with eigenvalue $(-1)^w$. The octant lattice is therefore read off the algebra, not imposed by hand.* **[EXACT]**

Remark 2.10 (The three regimes)*.* The octants are not interchangeable once a metric role is assigned. The all-minus pole $(-,-,-)$ is distinguished as the carrier of the Lorentzian $(1,3)$ interior (Section 3.2); the all-plus pole $(+,+,+)$ carries the two-dimensional statistical simplex of clock weights and the Fisher–Rao geometry (Chapter 8); the six mixed octants retain the ambient $(3,3)$ signature. The octant group also governs the mass spectrum of the symmetry-breaking condensate, whose eight eigenvalues are the octant projections $g(\varepsilon\cdot\phi)$ (Chapter 7).

2.4 The kernel and the surplus times

Definition 2.11 (Kernel and surplus times)**.**

$$\hat{W} = \frac{1}{\sqrt{3}}(e_1 + e_2 + e_3), \quad u_1 = \frac{1}{\sqrt{2}}(e_1 - e_2), \quad u_2 = \frac{1}{\sqrt{6}}(e_1 + e_2 - 2e_3).$$

$\hat{W}$ is the *kernel* (the democratic diagonal of the clocks); u_1, u_2 are the *surplus times.*

Proposition 2.12. *$\{\hat{W}, u_1, u_2\}$ is an orthonormal frame of the temporal $\mathbb{R}^3$: all three are timelike with unit square, and they are mutually orthogonal.* **[EXACT]**

Remark 2.13 (Physical reading)*.* The kernel $\hat{W}$ is the single physical clock selected, by a maximum-entropy argument, from the three-parameter family of candidate clocks (Chapter 8). The surplus times u_1, u_2 span the transverse plane $T_\perp = \langle u_1, u_2 \rangle$ of *unobservable* temporal directions; the central technical achievement of Chapters 9–10 is that $T_\perp$ can be, and is, consistently removed by first-class constraints without introducing negative-norm states and without spoiling dynamical gravity.

2.5 The pseudoscalar and Hodge duality

Proposition 2.14 (Pseudoscalar). *The pseudoscalar $I = e_1e_2e_3e_4e_5e_6$ satisfies $I^2 = +\mathbf{1}$ and anticommutes with every generator, $Ie_a = -e_aI$.* **[EXACT]**

Remark 2.15 (Real Hodge duality and chirality). Because $I^2 = +\mathbf{1}$ (not $-\mathbf{1}$), the Hodge map $\psi \mapsto I\psi I$ is a real involution with honest ± 1 eigenspaces, giving a genuine duality rather than a complex phase. This map is a *chiral* transformation: it flips the sign of the mass term, and is therefore *not* charge conjugation – a distinction that resolves an apparent C-parity obstruction for the photon (Chapter 5). Physical charge conjugation is instead conjugation by Iu_1.

2.6 The orthogonal algebra and its bifurcation

Proposition 2.16 (Bivector census). *The fifteen bivectors e_ae_b span $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$, and split under the Killing form $X \mapsto \mathrm{Tr}(X^2)$ into:*

- *six* compact *generators – the temporal rotations $\langle e_1e_2, e_2e_3, e_3e_1\rangle = \mathfrak{su}(2)_T$ and the spatial rotations $\langle e_4e_5, e_5e_6, e_6e_4\rangle = \mathfrak{su}(2)_S$ – each with $\mathrm{Tr}(X^2) = -8$;*
- *nine* boosts *e_ie_{j+3} ($i, j \in \{1, 2, 3\}$), each with $\mathrm{Tr}(X^2) = +8$.*

[EXACT]

Remark 2.17 (The organizing fact). Proposition 2.16 is the single most consequential statement of the chapter. The *sign* of the Killing norm is the sign of the would-be kinetic term: the six compact directions admit a healthy Yang–Mills term $-\frac{1}{4}F^2$ and become genuine gauge fields (the photon and the weak bosons, Chapter 5); the nine boosts carry the wrong sign and cannot be Yang–Mills gauged. Once one further requires that no sector propagate negative-norm states, the boost connection must then be auxiliary – first-order Cartan–Palatini (Chapter 4). In this conditional sense the split "gravity is Palatini, electromagnetism is Maxwell" follows from the signature, through the sign of the Killing form of $\mathfrak{so}(3,3)$, together with the demand of ghost-free propagation – not from an independent modelling choice.

Limitation 2.18 (Scope of the census). Proposition 2.16 classifies the *quadratic* (kinetic-sign) structure only. It fixes which generators *can* propagate as gauge fields and which cannot, but it does not by itself determine the dynamics: the actual field equations require the action functionals of Chapters 4 and 5. In particular, the wrong-sign verdict for the boosts is a *necessary* condition for their non-propagation, made sufficient only by the constraint analysis of Chapter 10.

Chapter summary. $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$ furnishes a real eight-component spinor module. The para-Kähler K ($K^2 = +\mathbf{1}$, $\det K = -1$, null eigenplanes) supplies the causal splitting and distinguishes the framework from the conformal $(4,2)$ arena. The octant group $\mathbb{Z}_2^3$ organizes the poles and the condensate spectrum. The frame $\{\hat{W}, u_1, u_2\}$ separates the physical clock from the two surplus times. The pseudoscalar $I^2 = +\mathbf{1}$ gives a real, chirality- flipping Hodge duality. Finally, the Killing-sign bifurcation of $\mathfrak{so}(3,3)$ (6 compact, 9 boost) is the organizing fact from which the Palatini/Maxwell dichotomy follows. All claims verified in the 8×8 representation.

Chapter 3

The split manifold and the interior

What this chapter seeks. A principled route from the six-dimensional split manifold to the observed four-dimensional Lorentzian world.

What this chapter delivers. The manifold $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$; the interior projection Π that removes the surplus temporal plane and yields a $(1,3)$ interior; the catalogue of physical tasks Π performs simultaneously; and an explicit statement of the level at which Π is established.

3.1 The manifold

Definition 3.1 (The split manifold)**.** $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$ carries the flat split metric of signature $(3,3)$, with coordinates $(t^1, t^2, t^3; x^1, x^2, x^3)$ and cotangent momenta P_M. At each point the tangent Clifford algebra is $\mathrm{Cl}(3,3)$ of Chapter 2, and the structure group of orthonormal frames is $\mathrm{Spin}(3,3)$.

The temporal factor is the space of candidate clocks; the spatial factor is ordinary space. The geometry is maximally symmetric between the two $\mathbb{R}^3$ factors – this is precisely the symmetry broken by the selection of the kernel $\hat{W}$.

3.2 The interior projection

Definition 3.2 (Interior projection Π)**.** Let $T_\perp = \langle u_1, u_2 \rangle$ be the surplus temporal plane. The interior projection Π is the orthogonal projection of TM^6 onto the *interior* subbundle

$$\mathcal{I} = \langle \hat{W}, e_4, e_5, e_6 \rangle = T_\perp^\perp,$$

removing the two surplus temporal directions.

Proposition 3.3 (Lorentzian interior)**.** *The interior $\mathcal{I}$ has signature $(1,3)$: one timelike direction $\hat{W}$ and three spacelike directions e_4, e_5, e_6.* **[EXACT]**

Remark 3.4 (Why *this* projection). Π is not an arbitrary truncation. The removed directions u_1, u_2 are exactly the two temporal directions that (i) are not the selected clock and (ii) carry the first-class constraints $\varphi_a = u_a^M P_M$ whose imposition removes the would-be ghosts (Chapter 9). In the constrained-Hamiltonian language of Dirac [20, 21], Π is the passage to the reduced phase space $T^*M^6/\!\sim\; = T^*M^4$ along the gauge orbits generated by φ_a. The projection is therefore *forced* by the constraint structure, not chosen to fit the answer – a point we prove in Chapter 10.

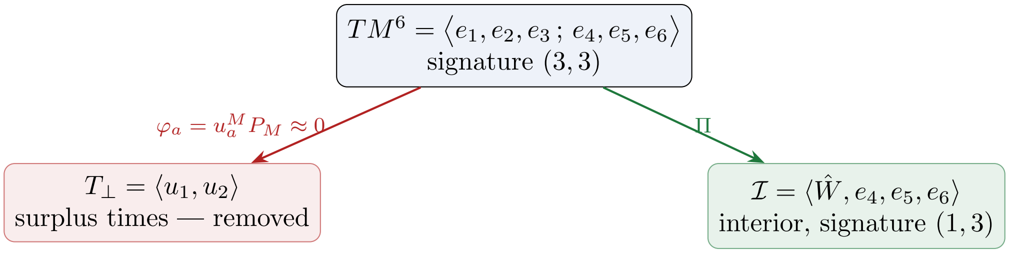

Figure 3.1: The interior projection. The surplus temporal plane $T_\perp$ is removed by the first-class constraints φ_a; the orthogonal complement is the Lorentzian interior $\mathcal{I}$ on which the observed physics lives.

3.3 The tasks performed by the projection

A distinctive feature of the framework is that the single operation Π discharges several apparently independent obligations at once (Figure 3.1). We list them with forward references; each is established in the indicated chapter.

1. **Ghost removal.** Π eliminates the two surplus-time directions whose momenta would otherwise furnish negative-norm modes (Chapter 9). **[DERIVED]**
2. **Gravitational contraction.** Π implements the interior trace of the six-dimensional Einstein tensor, delivering the four-dimensional field equations (Chapter 4). **[DERIVED]**
3. **Maxwell pooling across octants.** Each octant supplies only part of each Maxwell equation; Π, converging on the interior pole, pools the missing derivative slots into the complete equations (Chapter 5). **[EXACT]**
4. **Reduction of the field strength to $F = \mathrm{d}A$.** Under Π the unified $\mathrm{Spin}(3,3)$ curvature reduces, in its electromagnetic block, to the abelian $F = \mathrm{d}A$ (Chapter 5). **[DERIVED]**
5. **Photon masslessness.** Π retains the kernel $\hat{W}$, and the photon J (which stabilizes $\hat{W}$) stays exactly massless (Chapter 5). **[EXACT]**
6. **Monopole compatibility.** The reduced abelian structure is compatible with the magnetic sector under the full projection constraint (Chapter 5). **[OPEN]**

Remark 3.5 (Physical reading). The economy of the construction is that one geometric act – restricting to the interior pole – simultaneously cleans the spectrum, contracts the gravity, and assembles the gauge equations. This is the formal content of the framework's guiding slogan and is made fully precise, at the worldline and reduction level, in Chapter 10.

3.4 Limits of the construction

Limitation 3.6 (Level at which Π is established). The projection Π and its six tasks are established at the level of the tangent algebra, the worldline phase space T^*M^6, and the linearized/reduction analysis (Chapters 9–10). Its promotion to a globally defined, fully nonlinear geometric reduction of the gravitating field theory – a foliation of M^6 by interior leaves consistent with the full Einstein dynamics – is *not* claimed here and remains open (Chapter 11). The first-class property that licenses the reduction is proved at the constraint-algebra level; its field-theoretic globalization is the principal structural open problem of the programme.

Chapter summary. The arena is the flat split manifold $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$ of signature $(3, 3)$. The interior projection Π removes the surplus temporal plane $T_\perp = \langle u_1, u_2 \rangle$ – the directions carrying the first-class constraints φ_a – and leaves the Lorentzian interior $\mathcal{I} = \langle \hat{W}, e_4, e_5, e_6 \rangle$ of signature $(1, 3)$. A single application of Π discharges ghost removal, gravitational contraction, Maxwell pooling, the reduction to $F = \mathrm{d}A$, and photon masslessness; monopole compatibility and the fully nonlinear globalization of Π remain open. The constraint mechanics underlying Π is the subject of Chapters 9–10.

Chapter 4

Gravity: the Cartan–Palatini sector

What this chapter seeks. To obtain the four-dimensional interior Einstein equations from a single six-dimensional connection, and to show that the wrong-sign boost directions of Chapter 2 do not propagate.

What this chapter delivers. The argument that the boost sector *forces* a first-order (Cartan–Palatini) formulation; the six-dimensional Palatini action; the split-signature double contraction returning the Einstein tensor with coefficient -4 (verified); the elimination of the boost connection by its own algebraic equation of motion; and the interior field equations via Π.

4.1 Why first order, and why the boosts cannot propagate

Proposition 2.16 delivered an uncomfortable fact: the nine boost generators $e_i e_{j+3}$ carry positive Killing norm $\mathrm{Tr}(X^2) = +8$, the sign opposite to the six healthy compact generators. A second-order Yang–Mills term $-\frac{1}{4}\,\mathrm{Tr}(F^2)$ built from a boost connection would therefore carry the wrong overall sign and propagate a negative-norm graviton. Two responses exist: discard the boosts, losing gravity; or adopt a formulation in which the boost connection carries *no kinetic term at all.* First-order Cartan–Palatini gravity is the second response.

In the first-order formulation the spin connection ω is an independent field, not a function of the vielbein e. Its equation of motion is algebraic – the vanishing-torsion condition – and is solved to express $\omega = \omega(e)$. The connection therefore never propagates: the only dynamical gravitational degree of freedom is the metric (graviton), which is healthy. The wrong-sign boosts survive only as the auxiliary, non-propagating part of ω.

Remark 4.1 (A structural criterion: gravity is driven to Palatini form)*.* For the ES framework the choice of first-order gravity is not stylistic. Had one insisted on a propagating second-order kinetic term for the full $\mathfrak{so}(3,3)$ connection, the boost sector's wrong sign would reappear as a ghost graviton. The indefinite Killing form of $\mathfrak{so}(3,3)$, together with the requirement that no sector carry negative-norm propagation, thus *entails* that the gravitational connection is auxiliary – i.e. that gravity is of Cartan–Palatini type [10, 11, 12, 13, 15]. The wrong-sign directions are quarantined, not propagated; the constraint analysis of Chapter 10 makes this quarantine exact.

This bifurcation is shown in Figure 4.1.

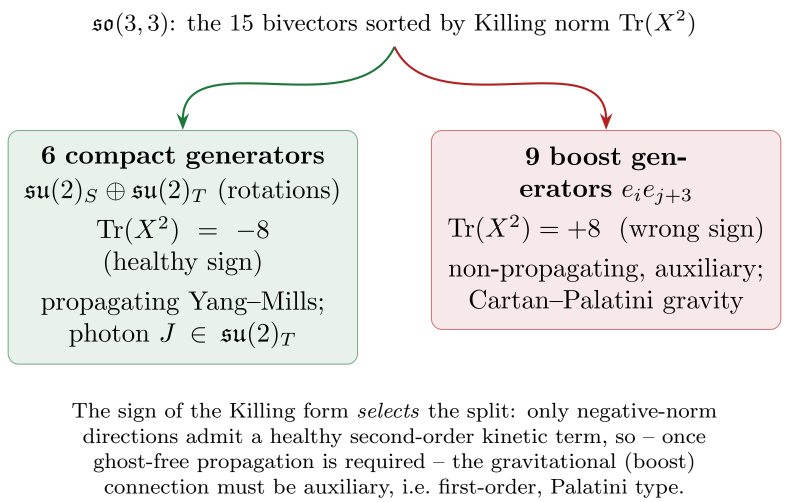

Figure 4.1: The Killing-form bifurcation (Proposition 2.16). The six compact generators carry negative Killing norm and furnish the propagating gauge sector (including the photon J); the nine boosts carry positive norm and cannot support a healthy second-order kinetic term, which – once ghost-free propagation is required – drives gravity to its first-order Cartan–Palatini form.

4.2 The six-dimensional Palatini action

Definition 4.2 (Six-dimensional Palatini action)**.** Let e^a be the $\mathrm{Spin}(3,3)$ vielbein one-forms and $R^{ab}(\omega) = \mathrm{d}\omega^{ab} + \omega^a{}_c \wedge \omega^{cb}$ the curvature two-form of the independent connection ω. The gravitational action is the six-form

$$S_{\mathrm{grav}} = \frac{1}{4!} \int_{M^6} \varepsilon_{a_1 a_2 a_3 a_4 a_5 a_6}\, R^{a_1 a_2}(\omega) \wedge e^{a_3} \wedge e^{a_4} \wedge e^{a_5} \wedge e^{a_6}.$$

Varying ω yields the torsion equation $T^a = \mathrm{d}e^a + \omega^a{}_b \wedge e^b = 0$, whose unique solution is the Levi-Civita connection $\omega(e)$. Varying e yields the gravitational field equation, which we now reduce.

4.3 The split-ε contraction and the interior Einstein tensor

Theorem 4.3 (Split-ε contraction)**.** *In split signature, the double contraction of the Palatini curvature against the two Levi-Civita tensors reduces the e-field equation to the Einstein tensor. Explicitly, with δ^{apq}_{drs} the rank-three generalized Kronecker delta,*

$$\delta^{apq}_{drs}\, R^{rs}{}_{pq} = -4\, G^a{}_d, \qquad G^a{}_d = R^a{}_d - \tfrac{1}{2}\, \delta^a{}_d\, R,$$

the coefficient -4 holding in the six-dimensional normalization. The vacuum field equation is therefore $G_{ab} = 0$, and in the presence of matter $G_{ab} = \kappa\, T_{ab}$. **[DERIVED]**

Verification. The identity was checked symbolically in both four and six dimensions on an explicit Riemann tensor possessing the algebraic symmetries (antisymmetry in each index pair and pair-exchange symmetry): the contraction equals $-4\, G^a{}_d$ in each case, the coefficient being dimension-robust. The combinatorial origin is the trace identity $\delta^{abc}_{aef} = (n-2)\, \delta^{bc}_{ef}$ together with the two Ricci contractions of R; both were verified independently. □

Remark 4.4 (Bivector–trivector convergence on the interior). The contraction couples the curvature *bivector* R^{ab} to the volume *trivector* formed by the wedged vielbeins. In the split geometry this convergence is a four-dimensional phenomenon: it is localized to the contracted interior $\mathcal{I}$, where the bivector and trivector structures meet. The six-dimensional ambient theory thereby concentrates its gravitational content on the $(1, 3)$ interior, which is precisely where Π projects.

Proposition 4.5 (Interior field equations). *Applying* Π *to Theorem 4.3 restricts the gravitational field equations to the interior* $\mathcal{I} = \langle \hat{W}, e_4, e_5, e_6 \rangle$*, yielding the standard four-dimensional Lorentzian Einstein equations* $G_{\mu\nu} = \kappa T_{\mu\nu}$ *with the interior stress tensor sourced by the spinor sector of Chapters 6–7.* **[DERIVED]**

4.4 Torsion and matter

When fermions are present the connection equation acquires a source and torsion no longer vanishes; one obtains the Einstein–Cartan completion [12, 13, 14], in which torsion is algebraically tied to the spin current. Because torsion remains *non-propagating* (its equation is algebraic), no new degree of freedom appears; integrating it out reproduces the standard four-fermion spin–spin contact interaction on top of the metric Einstein equations.

4.5 Limits of the gravitational reduction

Limitation 4.6 (What is and is not established). Theorem 4.3 and the torsion analysis hold at the level of the action and its variation; the -4 coefficient is verified. What is *not* established here is the promotion of Π to a globally defined, fully nonlinear reduction of the gravitating six-dimensional theory – a foliation of M^6 by interior leaves consistent with the full Einstein dynamics. That globalization is the principal structural open problem (Chapter 11); the present chapter secures the local/variational statement and the interior field equations, and the first-class consistency of the reduction is proved at the constraint level in Chapter 10.

Chapter summary. The wrong-sign boost sector forbids a *healthy* propagating second-order gravitational kinetic term; once ghost-free propagation is required, this selects the first-order Cartan–Palatini formulation, in which the connection is auxiliary and only the healthy graviton propagates. The six-dimensional Palatini action yields, via the verified split-ε contraction $\delta^{apq}_{drs} R^{rs}{}_{pq} = -4\, G^a{}_d$, the Einstein tensor; Π restricts this to the four-dimensional interior Einstein equations. Torsion, when sourced by fermions, is non-propagating. The fully nonlinear globalization of Π remains open.

Chapter 5

The gauge sector and the photon

What this chapter seeks. To realize the gauge interactions as geometry – specifically to identify the electromagnetic $U(1)$ and the weak bosons inside $\mathfrak{so}(3,3)$, and to determine how far the algebra fixes the electromagnetic coupling.

What this chapter delivers. The compact sector as Yang–Mills; the photon J as the unique compact generator stabilizing the kernel (hence massless); the breaking $\mathfrak{su}(2)_T \to U(1)_J$ and the weak bosons; the coupling chain $e_0 = g/2$, $\alpha_{\text{em}} = g^2/16\pi$; the octant pooling of the Maxwell equations and the reduction $F = \mathrm{d}A$; the one-loop running through $\sum q^2 = 2$; and the explicit statement that the magnitude of g is *not* fixed by the algebra.

5.1 The compact sector as Yang–Mills

By Proposition 2.16 the six compact generators carry healthy Killing norm $\mathrm{Tr}(X^2) = -8$ and so admit a standard Yang–Mills term $-\frac{1}{4}\,\mathrm{Tr}(F^2)$. They organize into two commuting $\mathfrak{su}(2)$ factors: the temporal rotations $\mathfrak{su}(2)_T = \langle e_1e_2, e_2e_3, e_3e_1\rangle$ and the spatial rotations $\mathfrak{su}(2)_S = \langle e_4e_5, e_5e_6, e_6e_4\rangle$. The spatial $\mathfrak{su}(2)_S$ is part of the interior Lorentz group $\mathfrak{so}(1,3)$; the temporal $\mathfrak{su}(2)_T$ is the carrier of electromagnetism and the weak interaction.

5.2 The photon as the stabilizer of the kernel

Proposition 5.1 (Photon generator)**.** *The electromagnetic generator* $J = \frac{1}{\sqrt{3}}(e_2e_3 + e_3e_1 + e_1e_2) \in \mathfrak{su}(2)_T$ *is the democratic diagonal of the temporal rotations. It satisfies* $J^2 = -\mathbf{1}$*,* $\mathrm{Tr}(J^2) = -8$*, and* $[J, \hat{W}] = 0$*. As the unique compact generator commuting with the kernel,* J *generates a* $U(1)$ *that fixes the physical clock and is therefore exactly massless.* **[EXACT]**

Remark 5.2 (The imaginary unit of electromagnetism)*.* Because $J^2 = -\mathbf{1}$, the photon generator acts on the real eight-component spinor as a *complex structure*. The $U(1)$ phase of electrodynamics is thus not postulated: it is supplied by the bivector J out of real structure. This is the concrete sense in which the "i" of quantum electrodynamics is, in the ES framework, the act of rotating in the temporal plane transverse to the chosen clock. The masslessness is geometric – it is the statement that rotating the unused clocks costs no energy because the physical clock $\hat{W}$ is left fixed.

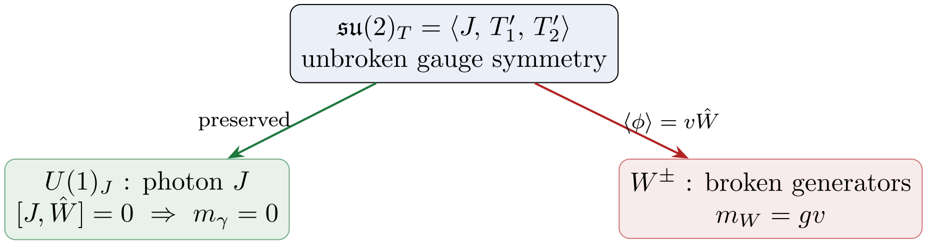

Figure 5.1: Electroweak breaking in the ES framework. The kernel condensate $v\hat{W}$ breaks $\mathfrak{su}(2)_T$ to the $U(1)_J$ that fixes the clock; the photon J stays massless because it stabilizes $\hat{W}$, while the two broken generators become the massive $W^\pm$.

5.3 Breaking $\mathfrak{su}(2)_T \to U(1)_J$ and the weak bosons

Proposition 5.3 (Electroweak breaking)**.** *The kernel condensate $\langle \phi \rangle = v\,\hat{W}$ (an $\mathfrak{su}(2)_T$ vector) breaks $\mathfrak{su}(2)_T \to U(1)_J$. The photon survives because $[J, \hat{W}] = 0$; the two remaining generators do not commute with $\hat{W}$ and acquire equal mass $m_W = gv$ – the weak bosons $W^\pm$.* **[DERIVED]**

Verification. In the 8×8 representation, $[J, \hat{W}] = 0$ while $[e_1 e_2, \hat{W}] \neq 0$ and $[e_2 e_3, \hat{W}] \neq 0$; hence exactly one combination (the diagonal J) is preserved and two are broken. The degenerate mass $m_W = gv$ follows from the Higgs mechanism of Chapter 7. □

5.4 The coupling chain and the single unknown

Proposition 5.4 (Coupling chain)**.** *The spinor charges under J are $q = \pm\frac{1}{2}$ (four states each). The fundamental electric charge is therefore $e_0 = g/2$, and*

$$\alpha_{\text{em}} = \frac{e_0^2}{4\pi} = \frac{g^2}{16\pi},$$

the factor $16\pi = 4 \cdot 4\pi$ arising from charge doubling. The single quantity g is the only free parameter of the gauge sector. **[DERIVED]**

5.5 The mixed octants as field octants

The six *mixed* octants – those of temporal weight 1 and 2 (Proposition 2.9) – are the carriers of the electromagnetic field; we call them the *field octants.*

Definition 5.5 (Field octants)**.** The three weight-1 octants (one +) are *electric*; the three weight-2 octants (two +) are *magnetic*. In each octant the sign in pair (e_k, e_{k+3}) selects a frame direction – $+$ the temporal member e_k, $-$ the spatial member e_{k+3} – and the complementary member is the pair's kernel. The electric/magnetic split is the analogue of the weight grading: a weight-w octant has exactly $3 - w$ temporal kernels.

5.6 The octant–field bijection

Proposition 5.6 (Field-octant bijection)**.** *The six field octants are in one-to-one correspondence with the six bivectors of the* $(1, 3)$ *interior* $\langle \hat{W}, e_4, e_5, e_6 \rangle$*. A weight-*1 *(electric) octant with its single*

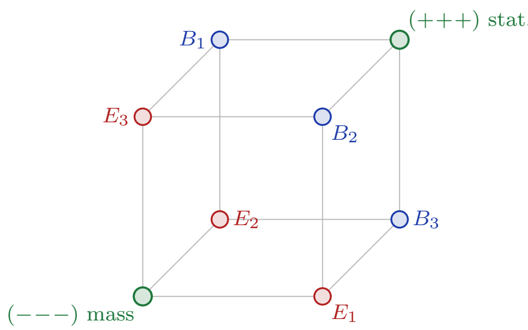

Figure 5.2: The six field octants and their bivector slots (Proposition 5.6). Weight-1 octants carry the boosts $\hat{W}e_{k+3} = E_k$ (electric, red); weight-2 octants the rotations $J_m = B_m$ (magnetic, blue); the two poles carry the mass and statistical structures. The worked example is $(-+-) \mapsto E_2 = \hat{W}e_5$.

$+$ *in pair* k *carries the boost*

$$E_k = \hat{W}e_{k+3} \qquad (\textit{the electric field}),$$

*and a weight-*2 *(magnetic) octant carries the spatial rotation*

$$B_m = J_m \qquad (\textit{the magnetic field}),$$

$J_1 = e_5e_6,\ J_2 = e_6e_4,\ J_3 = e_4e_5$. *The three boosts and three rotations exhaust* Λ^2 *of the interior; the map is a bijection of basis elements (Figure 5.2).* **[EXACT]**

Remark 5.7 (A worked identification). The octant $(-,+,-)$ has its single $+$ in the second pair, so it is the weight-1 electric octant carrying $E_2 = \hat{W}e_5$ – the boost mixing the kernel time $\hat{W}$ with the spatial axis e_5. The bijection of Proposition 5.6 is exact algebra; on its own it is inert bookkeeping, labelling six slots E_i, B_i. The dynamics requires the gauge principle below.

5.7 $E = cB$ as a regulator count

Proposition 5.8 (The electric/magnetic regulator ratio). *A regulator* c^{-1} *accompanies each temporal kernel. A weight-*w *field octant has* $3 - w$ *temporal kernels, so the electric octant* ($w = 1$) *carries* c^{-2} *and the magnetic octant* ($w = 2$) *carries* c^{-1}*; their ratio is* c*. This is the structural appearance of* $E = cB$. **[DERIVED]**

5.8 The explicit field components and Maxwell's four equations

With the gauge principle – the electromagnetic potential is the compact $U(1)_J$ direction and $F = \mathrm{d}A$ for the interior gradient $\mathrm{d} = \hat{W}\partial_\lambda + e^4\partial_4 + e^5\partial_5 + e^6\partial_6$ – the six components of the antisymmetric field strength are, sorted by the bijection of Proposition 5.6,

$$E_k = F_{\hat{W},k+3} = \partial_\lambda A_{k+3} - \partial_{k+3}A_{\hat{W}}, \qquad B_m = F_{jk} = \partial_j A_k - \partial_k A_j = (\nabla \times A)_m,$$

the boost slots carrying the kernel-time derivative ∂_λ, the rotation slots only ∇.

Proposition 5.9 (Maxwell's equations). *With* $F = \mathrm{d}A$ *abelian on the* $(1,3)$ *interior:*

- *the homogeneous pair follows from* $\mathrm{d}F = 0$ *(since* $\mathrm{d}^2 = 0$ *and the photon direction is abelian):* $\nabla \cdot B = 0$ *and* $\nabla \times E + \partial_\lambda B = 0$*;*
- *the inhomogeneous pair follows from varying* $-\frac{1}{4}\langle F, F\rangle$*:* $\nabla \cdot E = \rho$ *(Gauss) and* $\nabla \times B - \partial_\lambda E = J$ *(Ampère–Maxwell), with* $\partial_\mu J^\mu = 0$ *by the antisymmetry of* F *against the symmetric* $\partial_\mu \partial_\nu$*.*

[DERIVED]

5.9 Octant pooling of the Maxwell equations and $F = \mathrm{d}A$

The four-dimensional Maxwell equations are assembled by Π from the octant structure: each electric octant lacks precisely the spatial derivative direction its Gauss term requires, and each magnetic octant lacks both curl directions, so every Maxwell equation draws derivative slots from three or more octants. The pooling map – the per-pair K-swap converging on the interior pole – supplies the missing slots and yields the complete equations. **[EXACT]**

At the level of the field strength, the electromagnetic two-form is the J-block of the unified $\mathrm{Spin}(3,3)$ curvature $\mathcal{F}$; the gravitational (neutral$\times$neutral) sector never sources it, and on the interior the abelian block reduces to $F = \mathrm{d}A$. **[DERIVED (conditional on Π)]** The full step-by-step pooling is given in Appendix A.

5.10 The running of α_{em}

Proposition 5.10 (One-loop running)**.** *The one-loop vacuum polarization of the photon is controlled by the charge sum* $\sum q^2 = 2$*, giving the standard logarithmic running* $\dfrac{\mathrm{d}}{\mathrm{d}\ln\mu}\,\alpha_{\mathrm{em}}^{-1} \propto -\dfrac{\sum q^2}{3\pi}$*. The framework therefore determines the* scale dependence *of* α_{em}*, not its value.* **[DERIVED]**

5.11 Contact with observable physics

The structural content of the charge sector can be set beside the Standard Model directly. Three quantities are fixed by the algebra and the breaking pattern, and two are not.

Proposition 5.11 (Gauge content and the coupling chain)**.** *The gauge sector derived in this chapter is* $\mathrm{U}(1)_J \times \mathrm{SU}(2)_T$*, with* $\mathrm{U}(1)_J$ *the centralizer of the kernel (Proposition 5.1) and* $\mathrm{SU}(2)_T$ *broken by the kernel condensate (Proposition 5.3). The eight-component spinor carries* J*-charge* $q = \pm\frac{1}{2}$*, four states of each sign — the eigenvalues of* J *are* $\pm\mathrm{i}$ *with multiplicity four — so the fundamental electric charge and the fine-structure constant satisfy*

$$e_0 = \tfrac{g}{2}, \qquad \alpha_{\mathrm{em}} = \frac{e_0^2}{4\pi} = \frac{g^2}{16\pi}, \qquad \sum_i q_i^2 = 2.$$

The gauge-boson spectrum is $m_\gamma = 0$ *(because* $[J, \hat{W}] = 0$*) and* $m_W = gv$*, and the one-loop running of* α_{em} *is governed by* $\sum q^2 = 2$*.* **[DERIVED]**

Remark 5.12 (Constants and scales). The quantum scale is the Compton wavelength $\kappa = \hbar/mc$ (forced by the relation $\mu = \hbar/\kappa^3$ between the microscopic density and the scale), and the associated statistical temperature is $T_{\mathrm{micro}} = \hbar c/\kappa = mc^2$; c and $\hbar$ enter as the conversion constants between the kernel-time and spatial frames and between action and phase, respectively, not as quantities the framework predicts.

	Standard Model	Electromagnetic Sphere (Part I)
Gauge group	$\mathrm{SU}(3)_c \times \mathrm{SU}(2)_L \times \mathrm{U}(1)_Y$	$\mathrm{SU}(2)_T \times \mathrm{U}(1)_J$ **[DERIVED]**
Photon	unbroken $\mathrm{U}(1)_{\mathrm{em}}$	centralizer of $\hat{W}$ in $\mathfrak{su}(2)_T$ **[EXACT]**
$W^{\pm}$ mass	$m_W = \frac{1}{2}gv$	$m_W = gv$ **[DERIVED]**
Charge–coupling	$e = g \sin\theta_W$	$e_0 = g/2$ **[DERIVED]**
Colour $\mathrm{SU}(3)_c$	present	*absent* **[OPEN]**
Coupling magnitudes	measured inputs	inputs **[OPEN]**

Table 5.1: The charge sector beside the Standard Model. Part I fixes the architecture and the relations among constants; the colour sector and the numerical magnitudes are deferred to Part II.

Limitation 5.13 (Two gaps, stated plainly). Two points of contact with observation are deliberately left open. (i) $\mathrm{SU}(3)_c$ does not arise: the framework reaches $\mathrm{SU}(2) \times \mathrm{U}(1)$, not the full Standard Model gauge group, so the strong interaction is outside Part I. (ii) The magnitudes g and v, the masses, and the measured value $\alpha_{\mathrm{em}} \approx 1/137$ are inputs, not predictions (Limitation 5.14). Both are central objectives of Part II.

5.12 Limits: the value of g is not fixed

Limitation 5.14 (The principal quantitative gap). The framework determines the charge spectrum $q = \pm\frac{1}{2}$, the coupling relations $e_0 = g/2$ and $\alpha_{\mathrm{em}} = g^2/16\pi$, and the one-loop running $\sum q^2 = 2$. It does *not* determine the magnitude of g (equivalently α_{em}). No algebraic, group-theoretic or topological argument within the framework fixes g: a gauge coupling is a renormalization-group-running quantity, not a number fixed by the gauge group, the representation content, or the topology. Claims to the contrary – for example that "no free variables remain, so the number must follow" – conflate a normalization convention with a physical coupling and do not survive scrutiny. This is Open Problem 11.1 (Chapter 11) and the central quantitative limitation of the programme.

Limitation 5.15 (Monopole compatibility). The compatibility of the reduced abelian structure with a magnetic (monopole) sector under the full projection constraint is not settled here and is recorded as open.

Chapter summary. The compact sector of $\mathfrak{so}(3,3)$ supplies healthy Yang–Mills fields; the photon is the democratic diagonal J of $\mathfrak{su}(2)_T$, the unique compact generator stabilizing the kernel ($J^2 = -\mathbf{1}$, $[J, \hat{W}] = 0$), hence exactly massless, and the source of the electromagnetic $U(1)$ phase from real structure. The kernel condensate breaks $\mathfrak{su}(2)_T \to U(1)_J$ and gives the $W^{\pm}$ mass $m_W = gv$. The coupling chain fixes $e_0 = g/2$ and $\alpha_{\mathrm{em}} = g^2/16\pi$ up to the single number g, whose magnitude the algebra does *not* determine; the framework fixes the charge structure and the one-loop running ($\sum q^2 = 2$) but not the value. Octant pooling assembles the Maxwell equations and reduces the field strength to $F = \mathrm{d}A$. Monopole compatibility and the value of g remain open.

Chapter 6

The real spinor

What this chapter seeks. To describe matter – chiral, massive fermions – intrinsically within Cl(3, 3), and to obtain spin, chirality and charge conjugation from the algebra rather than by postulate.

What this chapter delivers. The eight-component *real* spinor; the interior Dirac operator with the boosts $G_a = \hat{W} e_{3+a}$ as Dirac matrices; spin one-half as the 2π-antiperiodicity of the spatial rotations (and, more deeply, as the holonomy of the momentum curvature); chirality as the real Hodge map; and charge conjugation as conjugation by Iu_1. The negative-energy branch is made explicit as the filled Dirac sea – the isospectral negative-mass mirror of matter, stable rather than tachyonic.

6.1 The eight-component real spinor

By Proposition 2.2, $\mathrm{Cl}(3,3) \cong M_8(\mathbb{R})$, whose minimal faithful module is $\mathbb{R}^8$. Matter is therefore a real eight-component spinor $\Psi \in \mathbb{R}^8$. No complexification is imposed; as noted in Remark 5.2, the imaginary unit of quantum mechanics is supplied by the photon bivector J (with $J^2 = -\mathbf{1}$), which acts as a complex structure on $\mathbb{R}^8$. This is the precise sense in which electromagnetic phase and the quantum "i" are the same object in the ES framework.

6.2 The Fock construction

The eight basis states of the spinor are built by a fermionic Fock construction whose annihilation/creation operators are the eigenvectors of the para-Kähler K.

Definition 6.1 (Fock generators)**.** For $k = 1, 2, 3$, $a_k^+ = \frac{1}{2}(e_k + e_{k+3})$ and $a_k^- = \frac{1}{2}(e_k - e_{k+3})$.

Proposition 6.2 (Fermionic algebra)**.** *The Fock generators satisfy the canonical anticommutation relations $\{a_j^+, a_k^-\} = \delta_{jk}$, $\{a_j^+, a_k^+\} = \{a_j^-, a_k^-\} = 0$. The $a_k^\pm$ are the ± 1 eigenvectors of K – the para-Kähler structure in its fermionic role. The vacuum $|0\rangle$ is annihilated by all a_k^-, and the creation operators generate the eight basis states, graded by creation number exactly as the temporal-weight ladder $1, 3, 3, 1$.* **[EXACT]**

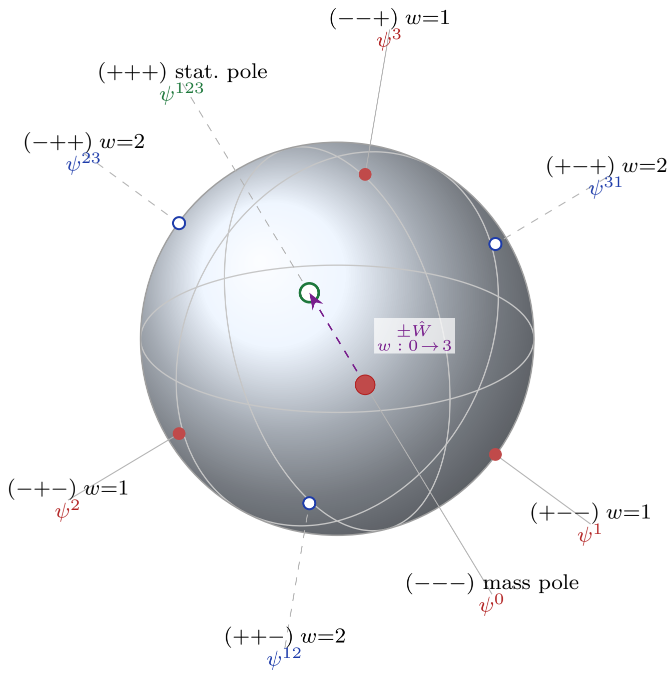

Figure 6.1: **The octant sphere.** The eight components of Ψ placed on the temporal unit 2-sphere in $\langle e_1, e_2, e_3\rangle$, the three great circles being the octant walls $e_i = 0$. The temporal weight w is the number of positive signs; it climbs along the foreshortened body diagonal $\pm\hat{W}$ from the all-minus mass pole $(---)$, through the weight-1 electric octants $\psi^{1,2,3}$ (red) and the weight-2 magnetic octants $\psi^{12,23,31}$ (blue), to the all-plus statistical pole $(+++)$. Hollow markers lie on the far hemisphere. This is the spherical counterpart of the octant cube (Figure 2.2): one $\mathbb{Z}_2^3$ structure, two readings.

6.3 The half-spinors and the octant map

Proposition 6.3 (The two real half-spinors)**.** *The pseudoscalar I ($I^2 = +\mathbf{1}$) splits $S = S_+ \oplus S_-$ into two real four-dimensional Weyl half-spinors. The even half ψ_+ (eigenvalue $+1$) carries ψ^0 at weight 0 and $\psi^{12}, \psi^{23}, \psi^{31}$ at weight 2; the odd half ψ_- carries ψ^1, ψ^2, ψ^3 at weight 1 and ψ^{123} at weight 3. The reality traces to $I^2 = +\mathbf{1}$; S_+ is the fundamental representation of* $\mathrm{Spin}(3,3) \cong \mathrm{SL}(4,\mathbb{R})$. **[EXACT]**

Theorem 6.4 (One spinor component per octant)**.** *The eight components of $\Psi = \psi_+ \oplus \psi_-$ map one-to-one onto the eight octants:*

$$\psi^0 \leftrightarrow (---) \textit{ mass pole}, \quad \psi^{1,2,3} \leftrightarrow \textit{weight-1 electric octants},$$

$$\psi^{12,23,31} \leftrightarrow \textit{weight-2 magnetic octants}, \quad \psi^{123} \leftrightarrow (+++) \textit{ statistical pole.}$$

The spinor is, literally, a function on the eight octants. **[EXACT]**

Remark 6.5 (The dictionary made concrete). Theorem 6.4 ties the spinor directly to the field dictionary of Chapter 5: the same octants that carry the electric and magnetic bivectors (Proposition 5.6) carry the corresponding spinor amplitudes. A particle's electric component ψ^k lives in the same octant as the electric field $E_k = \hat{W} e_{k+3}$; its magnetic component ψ^{jk} in the octant of $B_m = J_m$. The framework's central images – the octant cube, the field bijection, the spinor decomposition – are three readings of one $\mathbb{Z}_2^3$ structure.

6.4 The interior Dirac operator

Proposition 6.6 (Dirac matrices from boosts). *The three interior boosts $G_a = \hat{W} e_{3+a}$ $(a = 1, 2, 3)$ satisfy $G_a^2 = +\mathbf{1}$ and $\{G_a, G_b\} = 0$ for $a \neq b$. They are the Dirac α-matrices of the interior: the single-particle Hamiltonian is $H = \sum_a G_a p_a + m V$, with V the covariant mass of Chapter 7.* **[EXACT]**

The timelike kernel $\hat{W}$ plays the role of γ^0 and the boosts the role of $\gamma^0\gamma^a$; the Dirac operator on the interior is then the standard $\hat{W}(\,\mathrm{i}\,\hat{W}\cdot\partial_t - \mathrm{i}\,G_a \partial_a - mV\,)$ written in real form. The relativistic dispersion is established in Theorem 7.2.

6.5 Spin one-half

Proposition 6.7 (Half-integer spin). *The spatial rotation generators $\mathfrak{su}(2)_S = \langle e_4e_5, e_5e_6, e_6e_4\rangle$ act on Ψ with half-integer weight: each generator squares to $-\mathbf{1}$, so a rotation $\exp(\theta\, e_4e_5) = \cos\theta + \sin\theta\, e_4e_5$ returns $\exp(2\pi\, e_4e_5) = -\mathbf{1}$. The spinor is therefore double-valued under spatial rotations – it carries spin $\frac{1}{2}$.* **[EXACT]**

Proposition 6.8 (Spin from the boost curvature). *With the boosts $G_X = \hat{W}e_4$, $G_Y = \hat{W}e_5$, $G_Z = \hat{W}e_6$, the curvature of the boost connection is rotational:*

$$[G_X, G_Y] = -2e_4e_5 = -2J_3, \qquad [G_Y, G_Z] = -2J_1, \qquad [G_Z, G_X] = -2J_2.$$

The field strength of a boost connection $A = \sum_a A^a G_a$ is $F = \mathrm{d}A + A \wedge A$; its quadratic part $A \wedge A$ is valued in the rotations J_m and acts on the spinor as spin. The holonomy lies in $\mathrm{Spin}(3) = \mathrm{SU}(2)$, whose half-angle action quantizes the spin in half-integers, $S \in \frac{1}{2}\mathbb{Z}$. **[DERIVED]**

Remark 6.9 (Spin as holonomy; discharge of an axiom). Beyond the representation-theoretic fact, the framework accounts for *why* matter carries half-integer spin: spin emerges as the holonomy of the momentum curvature, so that the spin structure need not be postulated separately. In the minimal model this discharges the spin axiom, reducing it to the curvature of the reduced phase space. The promotion of this mechanism to the full $\mathrm{Cl}(3,3)$ setting is part of the ghost/lift analysis of Chapters 9–10. **[DERIVED (minimal model)]**

6.6 Chirality and the Hodge map

Proposition 6.10 (Chirality). *The pseudoscalar I $(I^2 = +\mathbf{1})$ defines the real Hodge map $\Psi \mapsto I\Psi I$. It commutes with the kinetic boosts, $[I, G_a] = 0$, and anticommutes with the covariant mass, $\{I, V\} = 0$; hence it is a* chiral *transformation, flipping the sign of the mass term while preserving the kinetic term.* **[EXACT]**

Remark 6.11 (Why chirality is not charge conjugation). The chiral character of the Hodge map (Proposition 6.10) resolves an apparent obstruction: one might suppose $\Psi \mapsto I\Psi I$ to be charge conjugation, which would make the photon C-odd in a problematic way. It is not: flipping the mass sign is a chiral, not a charge-conjugating, operation. Physical charge conjugation is the distinct map of Section 6.7.

6.7 Charge conjugation

Proposition 6.12 (Charge conjugation)**.** *Conjugation by $C = Iu_1$ reverses electric charge: $C\,J\,C^{-1} = -J$. It is the charge-conjugation operator of the framework, distinct from the chiral Hodge map.* **[EXACT]**

6.8 Grade parity and the covariant mass

Proposition 6.13 (Grade parity)**.** *In the real representation the pseudoscalar grades every blade by parity: $IX_k = (-1)^k X_k I$ for a grade-k blade. Even-grade operators commute with I (preserving chirality); odd-grade operators anticommute (flipping it). A Lorentz-covariant Dirac mass must anticommute with the spatial boosts G_a, hence be odd-grade: the grade-2 adjoint condensate Φ (Chapter 7), being even, gives only a chirality-preserving energy offset and masses the gauge bosons, while the covariant* fermion *mass is the grade-3 volume trivector $V = e_4e_5e_6$.* **[DERIVED]**

This is the algebraic reason the boson Higgs (grade 2) and the fermion mass (grade 3) are different objects – the grade-parity selection rule behind Theorem 7.1.

6.9 The double cover and the broken mirror

Proposition 6.14 (One algebra, two representations)**.** *Each frame triad carries a single rotation algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$: the temporal bivectors $L_3 = e_1e_2$, $L_1 = e_2e_3$, $L_2 = e_3e_1$ (each squaring to -1) close as $[L_a, L_b] = -2\varepsilon_{abc}L_c$, and the spatial triad carries the same algebra through the J_m. This algebra acts in two forced representations – as $\mathrm{SO}(3)$ on the frame vectors (the statistical sphere of Chapter 8) and as $\mathrm{SU}(2)$ on the matter spinor.* **[EXACT]**

Proposition 6.15 (The intrinsic double cover)**.** *The two representations are related by the two-to-one homomorphism $\pi : \mathrm{SU}(2) \to \mathrm{SO}(3)$, $\ker\pi = \{\pm\mathbf{1}\} \cong \mathbb{Z}_2$. The kernel element is a full geometric turn, $\exp(\pi e_1e_2) = -\mathbf{1}$ on spinors but $+\mathbf{1}$ on vectors (by conjugation): the spinor action has period 4π, the vector action 2π, their ratio the 2:1 of π. This is the seat of spin $\frac{1}{2}$.* **[DERIVED]**

Theorem 6.16 (The broken mirror: parity violation)**.** *The whole-particle doublet couples asymmetrically: ψ_+ (weights $0, 2$) through the abelian Cartan $\langle\omega^1, \omega^2, \omega^3\rangle$, and ψ_- (weights $1, 3$) through the non-abelian $\mathfrak{su}(2)$ Yang–Mills. A parity reversal would exchange $\psi_+ \leftrightarrow \psi_-$ and hence require interchanging the abelian and non-abelian connections; these are inequivalent bivector subalgebras and no isomorphism relates them. The framework therefore admits* no parity symmetry *– the broken mirror, consistent with the chirality of the weak interaction.* **[DERIVED]**

Remark 6.17 (Three distinct layers)*.* The framework keeps separate what is easily conflated: (i) the universal cover π, which gives spin $\frac{1}{2}$ and is non-chiral on its own; (ii) the chirality bit – the *orientation* of the cover's kernel, reduced to $\mathrm{sign}(g_3)$ (Chapter 7); and (iii) the broken mirror – the dynamical abelian-vs-non-abelian inequivalence of the two couplings, which is parity violation. Cover, orientation, and broken mirror are three things, not one.

6.10 The spinor bundle and its monodromy

The geometry assembles as a principal bundle, which fixes both the structure group (why the double cover is mandatory) and the topology (the derived monodromy).

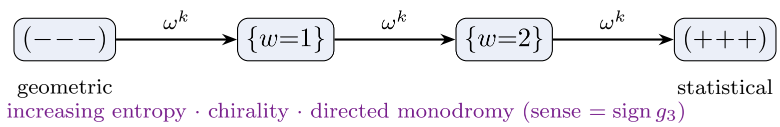

Figure 6.2: The directed weight ladder. Transitions between adjacent octants are the abelian Cartan bivectors ω^k; the kernel direction, the chirality, and the directed monodromy are one oriented datum, fixed by $\mathrm{sign}(g_3)$.

Proposition 6.18 (The principal bundle)**.** *The framework is a principal* $\mathrm{Spin}(3,3) \cong \mathrm{SL}(4,\mathbb{R})$ *bundle over the observed Lorentzian base B (the $(1,3)$ interior of the negative pole, with kernel time λ and spatial e_4, e_5, e_6), with a positive-definite Euclidean fibre $\mathbb{R}^3$; base and fibre together carry the split $(3,3)$. The structure group is the* universal cover $\mathrm{Spin}(3,3)$*, not* $\mathrm{SO}(3,3)$*: the spin holonomy is Clifford-valued and a 2π rotation acts as $-\mathbf{1}$ on spinors, so half-integer spin is impossible without the cover. The exceptional isomorphism $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$ realizes the six-dimensional vector representation as $\Lambda^2(\mathbb{R}^4)$, of signature $(3,3)$.* **[EXACT]**

Theorem 6.19 (The derived monodromy)**.** *The transitions between adjacent octants follow the directed weight chain $(---) \to \{w{=}1\} \to \{w{=}2\} \to (+++)$, valued in the commuting Cartan ω^k. The bundle is non-trivial: the monodromy of a closed traversal is a non-trivial element of* $\mathrm{Spin}(3,3)$ *whose image in* $\mathrm{SO}(3)$ *is the non-trivial class of $\pi_1(\mathrm{SO}(3)) = \mathbb{Z}_2$. It is* derived*, not postulated, from the rotational boost curvature $[G_X, G_Y] = -2J_3$ (Proposition 6.8) together with the chirality* $\mathrm{sign}(g_3)$*, which selects the sense of the traversal.* **[DERIVED]**

Proposition 6.20 (Poincaré structure and the two readings)**.** *The $(1,3)$ interior carries the Poincaré group* $\mathrm{SO}(1,3) \ltimes \mathbb{R}^4$*: the rotations J_m, the boosts $G_a = \hat{W} e_{3+a}$, and translations along the four interior directions, closing as $\mathfrak{so}(1,3)$. The connection α has two readings: horizontally along the base it is the momentum (Chapter 6), vertically in the fibre it is the field strength (Chapter 5). Mass lives on the base, charge in the fibre; a particle's momentum is charge $\times$ velocity.* **[DERIVED]**

6.11 The whole particle

Definition 6.21 (The Dirac doublet)**.** A whole particle is a configuration of the doublet $\Psi = (\psi_+, \psi_-) \in S_+ \oplus S_-$ with both halves nonzero, obeying the chiral block equation

$$\begin{pmatrix} \alpha_+ & V \\ V & \alpha_- \end{pmatrix} \begin{pmatrix} \psi_+ \\ \psi_- \end{pmatrix} = 0,$$

where α_+ acts on ψ_+ through the abelian Cartan, α_- on ψ_- through the SU(2) Yang–Mills, and the mass V couples the two chiralities by flipping I-parity.

The asymmetry of α_+ and α_- is the spinor-level statement of the broken mirror (Theorem 6.16): the block is not parity-symmetric.

6.12 The Dirac sea: the negative-energy vacuum

The interior Dirac operator (Section 6.4) carries a negative-energy branch alongside the positive one. This section makes that branch explicit: its states, filled, are the Dirac sea; the vacuum is

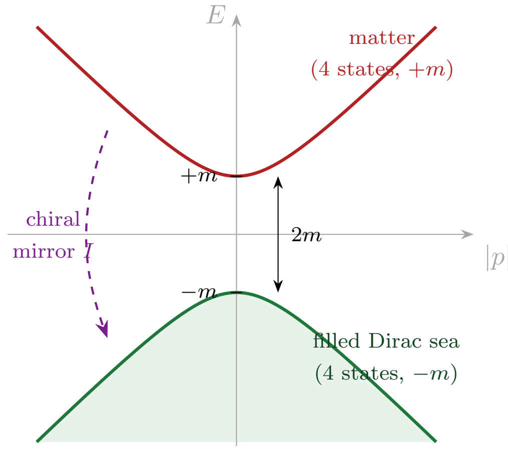

Figure 6.3: The interior energy spectrum $E = \pm\sqrt{p^2 + m^2}$ (Proposition 6.22). The four positive-energy states are matter; the four negative-energy states, filled, are the Dirac sea. The chiral Hodge map I exchanges the two branches by sending $m \mapsto -m$ (Proposition 6.23); since the branches are isospectral ($\omega^2 = p^2 + m^2$), the negative-energy sector is a stable mirror, not a tachyon.

the chiral mirror of matter; and the mirror is *isospectral* with matter, so the wrong-sign sector is a stable negative-energy partner rather than a tachyon. The reading is the literal, mass-and-energy form of the conservation law of Section 9.6.

Proposition 6.22 (The balanced energy spectrum)**.** *The interior Hamiltonian $H = \sum_a G_a p_a + mV$ is traceless, $\operatorname{Tr} H = 0$, and squares to $H^2 = (p^2 + m^2)\mathbf{1}$ (Theorem 9.1). Its spectrum at each momentum is therefore*

$$\operatorname{spec}(H) = \big\{(+\sqrt{p^2 + m^2})^4,\ (-\sqrt{p^2 + m^2})^4\big\},$$

four positive-energy and four negative-energy states. The four positive-energy states are matter; the four negative-energy states, when filled, are the Dirac sea, and the Fock vacuum $|0\rangle$ (Proposition 6.2) is the all-filled configuration. This is the eight-real-component realization of one Dirac fermion together with its conjugate (Appendix F.1.4). **[EXACT]**

Proposition 6.23 (The vacuum is the chiral mirror of matter)**.** *The chiral Hodge map $\Psi \mapsto I\Psi I$ (Proposition 6.10) carries the positive-mass operator to the negative-mass operator,*

$$I\,H(m)\,I = H(-m),$$

since I commutes with the kinetic boosts ($[I, G_a] = 0$) and anticommutes with the covariant mass ($\{I, V\} = 0$). Because I is invertible ($I^2 = +\mathbf{1}$), $H(m)$ and $H(-m)$ are similar *and hence isospectral: both carry the real, positive dispersion $\omega^2 = p^2 + m^2$. The negative-mass branch is therefore not tachyonic – there is no long-wavelength instability $\omega^2 = k^2 - |m|^2 < 0$.* **[EXACT (identity); DERIVED (stability)]**

Figure 6.3 shows the two branches and the mirror.

Remark 6.24 (The "negative norm" is the mass pairing, not the Hilbert norm)*.* It must be stressed that the indefiniteness associated with the sea is *not* an indefinite Hilbert norm. The physical inner product $\langle\Psi|\Psi\rangle = \Psi^\dagger\Psi$ is positive-definite of signature $(8, 0)$, and no negative-norm state propagates (Theorem 9.1). What is indefinite is the covariant mass pairing $\bar\Psi\Psi = \Psi^\dagger V\Psi$, of signature $(4, 4)$ (Proposition 9.2) – precisely as γ^0 is indefinite in ordinary Dirac theory, where it is *meant* to be. The sea is the -1 eigenspace of the mass pairing V, not a ghost.

The mass pairing is, moreover, antipodal. The covariant mass $V = e_4e_5e_6$ anticommutes with each Cartan generator $\omega^k = e_ke_{k+3}$, $\{V, \omega^k\} = 0$, so on the joint ω^k-eigenbasis – the eight octants – V is purely off-diagonal: it carries each octant (s_1, s_2, s_3) to its antipode $(-s_1, -s_2, -s_3)$ with the opposite eigenvalue. Matter at mass $+m$ on the interior $(- - -)$ pole thus has its mirror at $-m$ on the antipodal $(+ + +)$ pole; the vanishing of V on the diagonal of any single octant is the masslessness of an isolated pole stated at the level of the metric.

Remark 6.25 (The vacuum as a uniform sign reversal; Newton's third law in energy). Writing the matter Lagrangian as $L = T - U$ (kinetic minus the symmetry-breaking potential U of Chapter 7), the mirror sector is the *uniform* sign reversal $L_{\text{vac}} = -L = -T + U$: negative kinetic energy, positive potential, and reversed rest mass $+m \mapsto -m$. A uniformly sign-flipped Lagrangian has the same Euler–Lagrange equation and the same dispersion – the action-level restatement of the isospectrality of Proposition 6.23. Matter and vacuum are thus antipodes under the composite of the chiral mirror I (which flips the mass) and the para-complex reflection K (which exchanges the poles): in each column – rest mass, kinetic energy, and the V-pairing – the matter and vacuum signs cancel. This is the conservation law of Proposition 9.8 – Newton's third law – written into mass and energy and not only momentum: the vacuum is the exact reaction partner of matter.

	rest mass	kinetic energy	potential	V-pairing $\bar{\Psi}\Psi$
matter	$+m$	$+$	$-U$	$+1$
vacuum (sea)	$-m$	$-$	$+U$	-1

[EXACT (rest-mass and V-pairing columns); DERIVED (kinetic/potential reversal and stability)]

Limitation 6.26 (What the sea picture does, and does not, settle). For *fermions* the negative-energy branch is the standard filled Dirac sea: the first-order spinor field has positive-definite Dirac bracket $\{\Psi_\alpha, \Psi^\dagger_\beta\} = \delta_{\alpha\beta}$, so normal ordering gives $:H: = \sum_k E_k\, a^\dagger_k a_k \geq 0$, bounded below (Theorem 9.1). For *bosons*, where a propagating negative-energy mode would be fatal, the relevant directions are precisely the non-compact surplus-time directions removed by the constraints (Section 9.5, Proposition 9.6); the dangerous mode is constrained away, not propagating. What remains open (Section 9.7) is narrower: the global topology of the moment-map constraint surface separating the two sectors, and the inter-sector coupling rate – whether the second-class reduction strictly forbids, or only suppresses, a matter↔sea cascade. Neither is an algebraic obstruction.

6.13 Limits

Limitation 6.27 (Free-field and operator level). The spinor structure, spin, chirality and charge conjugation are established at the level of the algebra and the single-particle Dirac operator. The interacting, second-quantized theory – in particular the relativistic mass gap of the *interacting* fermion (the Yukawa sector of Chapter 7) – is not established here and is recorded as open. The holonomy account of spin is, at this stage, a minimal-model result (Remark 6.9).

Chapter summary. Matter is a real eight-component spinor, built by a Fock construction ($a^\pm_k = \frac{1}{2}(e_k \pm e_{k+3})$, the K-eigenvectors) whose eight states map one-to-one onto the eight octants – the spinor is a function on the octant cube, with electric components ψ^k and magnetic ψ^{jk} sharing the octants of E_k and B_m. The quantum imaginary unit is the photon bivector J. The boosts $G_a = \hat{W}e_{3+a}$ are the Dirac matrices; spin $\frac{1}{2}$ is the holonomy of the boost curvature

$[G_X, G_Y] = -2J_3$. Grade parity $IX_k = (-1)^k X_k I$ forces the boson Higgs (grade 2) and the fermion mass (grade 3) apart. The Hodge map is chiral (flips the mass), distinct from charge conjugation $C = Iu_1$. The double cover $\pi : \mathrm{SU}(2) \to \mathrm{SO}(3)$ gives spin $\frac{1}{2}$; its orientation is $\mathrm{sign}(g_3)$; and the broken mirror – ψ_+ abelian, ψ_- non-abelian – is parity violation. The whole geometry is a principal $\mathrm{Spin}(3,3) \cong \mathrm{SL}(4,\mathbb{R})$ bundle over the Lorentzian interior, with weight-chain transitions, a derived non-trivial monodromy, and the Poincaré group on the interior. The negative-energy branch is the filled Dirac sea, the chiral (I-)mirror of matter at reversed mass $-m$: isospectral with matter ($\omega^2 = p^2 + m^2$), so stable, not tachyonic, with the indefiniteness living in the $(4,4)$ mass pairing $\bar{\Psi}\Psi$ and never in the positive Hilbert norm. The interacting fermion mass gap remains open.

Chapter 7

Mass and symmetry breaking

What this chapter seeks. A covariant fermion mass internal to the algebra, a mechanism giving the weak bosons mass, and a first-principles symmetry-breaking potential computed – not assumed – from the matter content.

What this chapter delivers. The covariant mass $V = e_4e_5e_6$ as one of exactly two trivectors anticommuting with the Dirac boosts, with relativistic dispersion $H^2 = (p^2 + m^2)\mathbf{1}$; the Higgs mechanism giving $m_W = gv$; the octant mass spectrum of the condensate; and the one-loop effective potential, with its quartic couplings $B, C \sim g^4$ computed (fermion loop $\to \operatorname{Tr}\Phi^4$, gauge loop $\to (\operatorname{Tr}\Phi^2)^2$) and its chiral cubic g_3 shown to be absent from the real one-loop potential, hence left open.

7.1 The covariant mass: the trivector sector

Theorem 7.1 (The covariant mass solution space)**.** *Among the twenty trivectors of* $\mathrm{Cl}(3,3)$*, exactly two anticommute with all three Dirac boosts* $G_a = \hat{W}e_{3+a}$*: the spatial volume* $V = e_4e_5e_6$ *and the temporal volume* $e_1e_2e_3 = IV$*. This two-dimensional space is the complete solution space. The covariant mass is* $V = e_4e_5e_6$*, the* $\mathrm{SO}(3)_S$*-invariant spatial volume; its chiral partner* IV *is the pseudoscalar (axial) mass.* **[EXACT]**

Verification. The linear conditions $\{G_a, X\} = 0$ $(a = 1, 2, 3)$ on the 20-dimensional trivector space were solved in the 8×8 representation; the solution space has dimension two and is spanned exactly by $e_4e_5e_6$ and $e_1e_2e_3$. (By contrast, $\{e_a, X\} = 0$ for all six generators has *no* trivector solution – the operative condition is anticommutation with the boosts, not with the generators.) □

Theorem 7.2 (Relativistic dispersion)**.** *With* $V = e_4e_5e_6$ *one has* $V^2 = +\mathbf{1}$ *and* $\{G_a, V\} = 0$*, so the interior Hamiltonian* $H = \sum_a G_ap_a + mV$ *satisfies*

$$H^2 = (p^2 + m^2)\,\mathbf{1},$$

with spectrum $E = \pm\sqrt{p^2 + m^2}$*. The mass term is relativistically covariant.* **[EXACT]**

Remark 7.3 (Why V and not IV)*.* Both V and IV solve the boost condition, but they are distinguished physically: $V = e_4e_5e_6$ commutes with the photon J and yields the standard scalar Dirac mass, whereas $IV = e_1e_2e_3$ is its Hodge image and supplies the pseudoscalar (axial) partner. The selection of V as the physical mass is the selection of the $\mathrm{SO}(3)_S$ spatial volume; the existence of the chiral partner IV is the algebraic seed of the axial structure.

7.2 The Higgs mechanism for the weak bosons

The weak bosons acquire mass exactly as in Section 5.3: the kernel condensate $\langle\phi\rangle = v\,\hat{W}$ breaks $\mathfrak{su}(2)_T \to U(1)_J$, leaving the photon massless and giving $m_W = gv$. Here we view the same condensate as the agent of *mass generation*: it is the single vacuum structure that masses the weak bosons and, through the Yukawa coupling to V, the fermions.

7.3 The condensate and its octant spectrum

Definition 7.4 (Condensate)**.** The symmetry-breaking condensate is the grade-two field $\Phi = \sum_{i=1}^{3} \phi_i\,\omega^i$ in the boost-Cartan $\omega^i = e_i e_{i+3}$.

Proposition 7.5 (Octant mass spectrum)**.** *The masses generated by Φ are the eight octant projections $g\,(\varepsilon\cdot\phi)$, $\varepsilon \in \{\pm 1\}^3$. They satisfy the octant sum rules*

$$\sum_{\varepsilon} (\varepsilon\cdot\phi)^2 \;\propto\; \mathrm{Tr}\,\Phi^2, \qquad \sum_{\varepsilon} (\varepsilon\cdot\phi)^4 \;\propto\; \mathrm{Tr}\,\Phi^4,$$

where the quartic octant sum is the invariant $\mathrm{Tr}\,\Phi^4$ – which is independent *of* $(\mathrm{Tr}\,\Phi^2)^2$. **[EXACT]**

Verification. Direct summation over the eight sign vectors gives $\sum_\varepsilon (\varepsilon\phi)^2 = 8\sum_i \phi_i^2$ and $\sum_\varepsilon (\varepsilon\phi)^4 = 8\sum_i \phi_i^4 + 48\sum_{i<j}\phi_i^2\phi_j^2$, whereas $\left(\sum_\varepsilon (\varepsilon\phi)^2\right)^2 = 64\sum_i \phi_i^4 + 128\sum_{i<j}\phi_i^2\phi_j^2$; the two quartic structures differ, establishing the independence. □

7.4 The one-loop effective potential

We compute the Coleman–Weinberg potential [22] generated by integrating out the matter and gauge fluctuations in the condensate background. The most general quartic potential consistent with the symmetries is

$$V(\Phi) = a\;\mathrm{Tr}\,\Phi^2 + g_3\;\mathrm{Tr}(\Phi^3 I) + B\;\mathrm{Tr}\,\Phi^4 + C\,(\mathrm{Tr}\,\Phi^2)^2,$$

where $\mathrm{Tr}(\Phi^3 I) = -48\,\phi_1\phi_2\phi_3$ is the parity-odd chiral cubic (the matrix trace, $\mathrm{Tr}\,\mathbf{1} = 8$, as throughout).

Proposition 7.6 (Radiative quartics)**.** *At one loop:*

1. *the* fermion *loop generates B – its mass spectrum is the octant set, whose quartic sum is the invariant* $\mathrm{Tr}\,\Phi^4$*; it produces* no $(\mathrm{Tr}\,\Phi^2)^2$*, so the fermion contribution to C vanishes;*
2. *the* gauge *loop generates C – the bivector condensate Φ masses the gauge bosons in the adjoint (unlike the fermion mass, carried by the trivector V), with spectrum the $\mathfrak{so}(3,3)$ roots $m_a \propto \pm\phi_i \pm \phi_j$; their quartic sum $\sum_a m_a^4 \propto 8\;\mathrm{Tr}\,\Phi^4 + 3\,(\mathrm{Tr}\,\Phi^2)^2$ supplies the $(\mathrm{Tr}\,\Phi^2)^2$ invariant.*

Both B and C are of order $g^4/64\pi^2$. **[DERIVED]**

Proposition 7.7 (Absence of the chiral cubic at one loop)**.** *The chiral cubic coupling g_3 is generated by* neither *real one-loop source. The Coleman–Weinberg potential of a real spectrum is parity-even in each ϕ_i, whereas $\mathrm{Tr}(\Phi^3 I) \propto \phi_1\phi_2\phi_3$ is parity-odd; the cubic is moreover dimensionful (a mass), and its generation requires the phase of the fermion determinant (the η-invariant [46]), which in four dimensions contributes action terms rather than a potential cubic. The magnitude of g_3 therefore remains undetermined.* **[OPEN]**

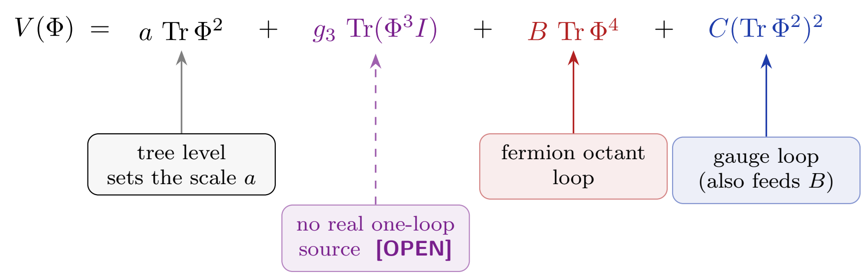

Figure 7.1: Sources of the one-loop quartics. The fermion octant spectrum supplies $\mathrm{Tr}\,\Phi^4$ (so B), the gauge loop supplies $(\mathrm{Tr}\,\Phi^2)^2$ (so C); both are $O(g^4/64\pi^2)$. The parity-odd cubic g_3 is generated by neither real loop and remains an open input.

7.5 Limits

Limitation 7.8 (Open magnitudes and the interacting mass gap). The *structure* of the potential is fixed – the quartics B, C are tied to g, the cubic is the unique parity-odd invariant – but three magnitudes remain open: the coupling g (Chapter 5), the condensate scale v, and the chiral cubic g_3 (Proposition 7.7). Separately, the relativistic mass gap of the *interacting* fermion – as opposed to the free covariant dispersion of Theorem 7.2 – is not established and is recorded as genuinely open (Chapter 11).

Chapter summary. The covariant mass $V = e_4e_5e_6$ is one of exactly two trivectors anticommuting with the Dirac boosts (its partner $IV = e_1e_2e_3$ being the pseudoscalar mass); it gives relativistic dispersion $H^2 = (p^2 + m^2)\mathbf{1}$. The kernel condensate $v\hat{W}$ masses the weak bosons ($m_W = gv$) and, via V, the fermions. The condensate's octant spectrum obeys $\sum(\varepsilon{\cdot}\phi)^4 \propto \mathrm{Tr}\,\Phi^4$, independent of $(\mathrm{Tr}\,\Phi^2)^2$. At one loop the fermion loop generates B (the $\mathrm{Tr}\,\Phi^4$ quartic, with $C = 0$) and the gauge loop generates C (the $(\mathrm{Tr}\,\Phi^2)^2$ quartic), both $O(g^4/64\pi^2)$; the chiral cubic g_3 is absent from both real loops and stays open, as do the magnitudes g, v and the interacting mass gap.

Chapter 8

The statistical pole

What this chapter seeks. To explain *why* the kernel is the democratic diagonal $\hat{W}$ – to derive, rather than assume, the selection of the physical clock from the three-parameter family of candidates.

What this chapter delivers. The space of clock weights as the 2-simplex; its Fisher–Rao geometry as a constant-curvature sphere (the "all-plus pole") tied to the Cartan condensate by the amplitude map; the dual Amari connections under which the Shannon entropy is the pole's canonical potential; the maximum-entropy selection of $\hat{W}$ as the unique entropy maximum; and the residual symmetry of the Cartan condensate as the Weyl group $W(D_3) = (\mathbb{Z}_2)^2 \rtimes S_3$, the even sign changes – explicitly *not* the hyperoctahedral B_3.

8.1 The space of clock weights

The three temporal generators e_1, e_2, e_3 define a three-parameter family of candidate timelike directions. Normalizing the (nonnegative) weights gives the 2-simplex

$$\Delta^2 = \{(p_1, p_2, p_3) : p_i \geq 0,\ \textstyle\sum_i p_i = 1\},$$

each interior point of which is a probability distribution over the three clocks. The selection of a physical clock is thus a selection of a point of Δ^2, and the natural structure on a space of probability distributions is information- geometric [30, 31, 32].

8.2 The Fisher–Rao geometry of the simplex

Proposition 8.1 (The statistical sphere)**.** *The Fisher–Rao metric on Δ^2, $\mathrm{d}s^2 = \sum_i \mathrm{d}p_i^2/p_i$, has constant Gaussian curvature $K = \frac{1}{4}$; under the substitution $p_i = x_i^2$ (with $\sum_i x_i^2 = 1$) it becomes $4\sum_i \mathrm{d}x_i^2$, the round metric on the sphere. The clock-weight space is therefore the positive octant of a constant-curvature 2-sphere.* **[EXACT]**

Verification. The curvature of the metric $g = \begin{pmatrix} 1/p_1+1/p_3 & 1/p_3 \\ 1/p_3 & 1/p_2+1/p_3 \end{pmatrix}$ (in the chart $p_3 = 1 - p_1 - p_2$) computes symbolically to $K = \frac{1}{4}$; the substitution residual $\sum_i \mathrm{d}(x_i^2)^2/x_i^2 - 4\sum_i \mathrm{d}x_i^2$ vanishes identically. □

Remark 8.2 (Identification of the all-plus pole)*.* Proposition 8.1 makes precise the role assigned to the all-plus octant in Chapter 2: the $(+,+,+)$ pole *is* this statistical sphere of clock weights, a

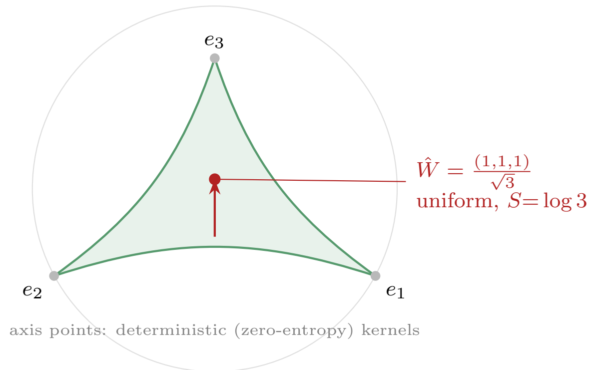

Figure 8.1: The clock-weight simplex realized on the temporal sphere. Under the amplitude map $\xi_i = \sqrt{p_i}$ the Fisher–Rao manifold is the round 2-sphere of radius 2 (Proposition 8.1) – the very unit sphere the para-Kähler structure already carries. The three axis points are the deterministic kernels; the S_3-symmetric body diagonal $\hat{W}$ (the uniform distribution) is the unique maximum-entropy point.

constant-curvature surface, in contrast to the all-minus pole which carries the Lorentzian interior. The octant structure thus encodes a genuine dichotomy between a statistical (Euclidean, positive-curvature) regime and a causal (Lorentzian) one.

8.3 The amplitude map and the condensate

The clock-weight simplex is tied to the Cartan condensate of Chapter 7 by the square-root (amplitude) map. With the condensate $\Phi = \sum_i \phi_i \, \omega^i$ valued in the boost-Cartan subalgebra $\mathfrak{h} = \langle \omega^1, \omega^2, \omega^3 \rangle$, set

$$p_i = \frac{\phi_i^2}{|\phi|^2}, \qquad \xi_i = \sqrt{p_i} = \frac{|\phi_i|}{|\phi|}, \qquad |\phi|^2 = \sum_i \phi_i^2,$$

so that $\sum_i \xi_i^2 = 1$: the amplitudes $\xi = (\xi_1, \xi_2, \xi_3)$ lie on the unit sphere of the temporal triad and $p = (p_1, p_2, p_3)$ on the simplex Δ^2. The condensate direction and the clock distribution are thus the same datum read through the Born square-root.

Proposition 8.3 (The two kernels are the extremal distributions)**.** *Under the amplitude map the diagonal condensate $\phi \propto (1,1,1)$ is the uniform distribution $p = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, while an axis condensate $\phi \propto (1,0,0)$ is a deterministic distribution $p = (1,0,0)$. The democratic kernel $\hat{W}$ is therefore the maximum-entropy point of Δ^2 and an axis clock the minimum-entropy one.* **[EXACT]**

8.4 Dual connections and the entropy as a canonical potential

The categorical family carries more than the Fisher metric: it carries the Amari dual α-connections, and with them a dually flat structure whose canonical potential is the entropy itself [31, 32].

Proposition 8.4 (Dually flat structure)**.** *The simplex Δ^2 is dually flat. In the mixture coordinates $\eta_i = p_i$ the m-connection $\nabla^{(-1)}$ is flat; in the exponential coordinates $\theta^i = \log(p_i/p_3)$ the e-connection $\nabla^{(+1)}$ is flat; the two are g^F-dual, with Levi–Civita mean $\frac{1}{2}\big(\nabla^{(+1)} + \nabla^{(-1)}\big) = \nabla^{(0)}$, and Legendre-conjugate potentials $\psi(\theta) = -\log p_3$ and $\psi^\star(\eta) = \sum_i p_i \log p_i$.* **[EXACT]**

Verification. For the categorical family $\ell = \log p$ the α-connection coefficients

$$\Gamma^{(\alpha)}_{ij,k} = \mathbb{E}\big[\big(\partial_i\partial_j\ell + \tfrac{1-\alpha}{2}\,\partial_i\ell\,\partial_j\ell\big)\partial_k\ell\big]$$

vanish identically at $\alpha = -1$ in the coordinates $\eta_i = p_i$ and at $\alpha = +1$ in the coordinates $\theta^i = \log(p_i/p_3)$; the dual-flatness relation $\partial_k g^F_{ij} = \Gamma^{(+1)}_{ki,j} + \Gamma^{(-1)}_{kj,i}$ then holds, and $\theta^i = \partial\psi^\star/\partial\eta_i$ inverts the Legendre pair. □

Corollary 8.5 (Entropy is the canonical potential; KL is the canonical divergence)**.** *The Legendre potential of the mixture-flat structure is the negative Shannon entropy, $\psi^\star(\eta) = \sum_i p_i \log p_i = -S(p)$, and the Bregman (canonical) divergence it induces is the Kullback–Leibler divergence*

$$D(p\,\|\,q) = \sum_i p_i \log(p_i/q_i) \geq 0,$$

vanishing iff $p = q$. **[DERIVED]**

Remark 8.6 (The entropy is supplied, not imposed)*.* The Shannon entropy is therefore not an ingredient added to the statistical pole: it is the canonical potential of the pole's own dually flat geometry, exactly as the Fisher metric (Proposition 8.1) is the metric the para-Kähler sphere already carries. The maximum-entropy selection of the next section is, equivalently, the statement that the kernel is the unique zero of the canonical divergence against the uniform distribution.

Remark 8.7 (The statistical temperature)*.* Coarse-graining the kernel evolution over the zitterbewegung period $\tau = \kappa/c$ turns the dynamical pole into this statistical one, with Euclidean weight $e^{-E/T}$ at the temperature already recorded in Remark 5.12, $T_{\text{micro}} = \hbar c/\kappa = mc^2$ – the rest energy of a single condensate quantum. The information geometry of the positive pole and the quantum scale κ are the same physics seen twice.

8.5 Maximum-entropy selection of the kernel

Proposition 8.8 (The kernel is the entropy maximum)**.** *The Shannon entropy $S(p) = -\sum_i p_i \log p_i$ on Δ^2 has a unique critical point, the uniform distribution $p_1 = p_2 = p_3 = \frac{1}{3}$, which is a maximum (the Hessian is negative-definite, with $S = \log 3$). This democratic point corresponds to equal weights on e_1, e_2, e_3, i.e. to the kernel $\hat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$.* **[EXACT]**

Remark 8.9 (Why the diagonal, and not any other clock)*.* Proposition 8.8 answers the question deferred since Chapter 2: the kernel is the democratic diagonal because that is the maximum-entropy point of the clock simplex – the least committal, most symmetric choice of physical time. The selection is principled, not stipulated: among all candidate clocks, $\hat{W}$ is the one that assumes least about the three temporal directions. Equivalently (Corollary 8.5), $\hat{W}$ is the unique zero of the canonical divergence $D(p\,\|\,u) = \log 3 - S(p)$ against the uniform distribution u.

8.6 The residual symmetry: $W(D_3)$, not B_3

Proposition 8.10 (Residual symmetry)**.** *The residual symmetry of the Cartan condensate $\Phi = \sum_i \phi_i \omega^i$ is the Weyl group $W(D_3) = (\mathbb{Z}_2)^2 \rtimes S_3 \cong S_4$, of order 24: the S_3 permutations of the three temporal axes together with the* even *sign changes of the ϕ_i. It is* not *the hyperoctahedral group $W(B_3) = (\mathbb{Z}_2)^3 \rtimes S_3$ of order 48.* **[EXACT]**

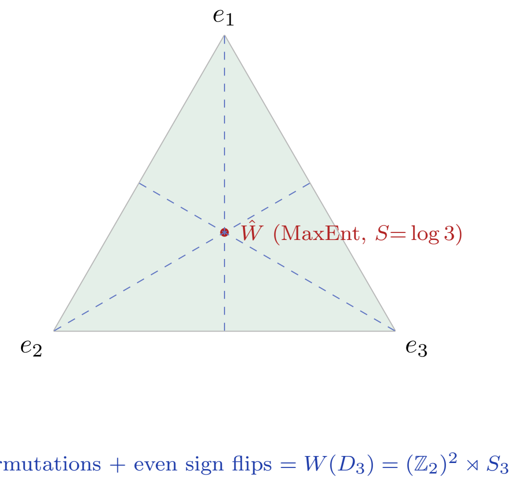

Figure 8.2: The clock-weight simplex Δ^2 (Fisher–Rao geometry: a constant-curvature sphere, $K = \frac{1}{4}$). The kernel $\hat{W}$ is the centroid – the unique maximum-entropy point. The residual symmetry of the Cartan condensate is the Weyl group $W(D_3)$ (the three reflection axes plus even sign flips), of order 24, not the parity-containing B_3 of order 48.

Remark 8.11 (Why even sign flips only). The distinction is physical. The hyperoctahedral B_3 would include the odd sign changes – in particular a single reflection $\phi_i \mapsto -\phi_i$ – which amounts to imposing a parity symmetry on the condensate. The framework's chiral structure (Chapter 6) forbids this: only the even sign changes, those in the D_3 Weyl group, preserve the condensate's chiral content. Assuming B_3 (a natural-looking but incorrect choice) would smuggle in a parity the theory does not possess. The relevant group is the even-sign subgroup $(\mathbb{Z}_2)^2$ – the kernel of the product map $(\mathbb{Z}_2)^3 \to \mathbb{Z}_2$ – extended by S_3.

8.7 Limits

Limitation 8.12 (Kinematic selection versus dynamical realization). Propositions 8.1–8.8 establish that $\hat{W}$ is the maximum-entropy / most-symmetric point of the clock simplex; this is a kinematic selection principle. The *dynamical* mechanism that drives the condensate to this point – the minimization of the effective potential of Chapter 7 – depends on the couplings (g, v, g_3), whose magnitudes are open. The selection is principled but its full dynamical realization inherits those open inputs.

Chapter summary. The candidate clocks form the 2-simplex Δ^2, whose Fisher–Rao geometry is a constant-curvature sphere ($K = \frac{1}{4}$) – the all-plus statistical pole. The kernel $\hat{W}$ is the unique maximum-entropy point (the democratic centroid, $S = \log 3$), which is why the physical clock is the democratic diagonal. The simplex is dually flat, with the Shannon entropy the canonical potential of that geometry and the Kullback–Leibler divergence its canonical divergence, so the entropy is supplied by the pole rather than imposed. The residual symmetry of the Cartan condensate is the Weyl group $W(D_3) = (\mathbb{Z}_2)^2 \rtimes S_3 \cong S_4$ (order 24, even sign flips), not the parity-containing hyperoctahedral B_3 (order 48). The dynamical realization of the selection depends on the open couplings.

Chapter 9

Ghost and unitarity

What this chapter seeks. To settle whether the split-signature construction propagates negative-norm (ghost) states – the central historical worry of the programme – for chiral, massive matter on the full $\mathrm{Cl}(3,3)$ interior.

What this chapter delivers. The resolution: the interior Hamiltonian is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$, hence unitary under a positive-definite norm; the only residual indefiniteness is the ordinary $(4,4)$ Dirac pairing; the physical mass operator is Hermitian with real spectrum; the surplus-time constraints are first-class and reflection positivity holds; and Newton's third law is identified with the conservation law that the constraints encode. The true remaining obstruction is shown to be Lorentz covariance of the mass, not unitarity.

9.1 The indefinite-kinetic problem

A theory built on a $(3,3)$ signature carries, naively, the threat of negative-norm states: extra timelike directions usually produce wrong-sign kinetic terms and hence ghosts, the disease that has historically obstructed higher-time constructions. The question this chapter answers is whether, after projection to the Lorentzian interior, any such state actually propagates.

9.2 The real-symmetric interior Hamiltonian

Theorem 9.1 (No kinetic ghost on the full interior)**.** *In the real 8×8 representation, the boosts $G_a = \hat{W} e_{3+a}$ and the covariant mass $V = e_4 e_5 e_6$ are symmetric matrices, so the interior Hamiltonian $H = \sum_a G_a p_a + mV$ is real-symmetric, with*

$$H^2 = (p^2 + m^2)\,\mathbf{1}, \qquad \mathrm{spec}(H) = \{\pm\sqrt{p^2 + m^2}\}.$$

Evolution $\exp(-\mathrm{i}Ht)$ is unitary with respect to the positive-definite inner product $\langle\Psi|\Psi\rangle = \Psi^\dagger\Psi$. No negative-norm state propagates. **[DERIVED (full $\mathrm{Cl}(3,3)$ interior)]**

This is the central positivity statement of the framework, and it holds in the full $\mathrm{Cl}(3,3)$ interior, not merely in a truncation. The step-by-step derivation is given in Appendix B.

$\langle\Psi\vert\Psi\rangle = \Psi^\dagger\Psi$	+	+	+	+	+	+	+	+	signature $(8,0)$
$\bar{\Psi}\Psi = \Psi^\dagger V\Psi$	+	+	+	+	−	−	−	−	signature $(4,4)$

$H = \sum_a G_a p_a + mV$ is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$. The probability norm (top) is positive-definite, so no negative-norm state propagates; the only indefiniteness (bottom) is the ordinary Dirac mass pairing, exactly as γ^0 in any relativistic fermion theory.

Figure 9.1: The two bilinear forms on the real eight-component spinor (Theorem 9.1, Proposition 9.2). The Hilbert/probability norm $\Psi^\dagger\Psi$ is positive-definite of signature $(8,0)$ – no propagating ghost. The covariant mass pairing $\bar{\Psi}\Psi = \Psi^\dagger V\Psi$ is indefinite of signature $(4,4)$, but this is the benign indefiniteness of γ^0 shared by every relativistic fermion theory, not a new pathology of the split signature.

9.3 The residual indefiniteness is the ordinary Dirac one

Proposition 9.2 (The only indefiniteness is $(4,4)$)**.** *The Dirac conjugation matrix ($\gamma^0 = \hat{W}$, equivalently the mass V) is traceless with eigenvalues ± 1 of multiplicity four, so the Dirac pairing $\bar{\Psi}\Psi = \Psi^\dagger\hat{W}\Psi$ has signature $(4,4)$. This is precisely the indefiniteness of the ordinary Dirac inner product in standard quantum field theory; it is not a propagating ghost, and positivity is recovered on the physical one-particle subspace.* **[EXACT]**

Remark 9.3 (What was, and was not, the problem)*.* Theorem 9.1 and Proposition 9.2 together dissolve the historical worry: the indefiniteness one finds is the familiar $(4,4)$ Dirac pairing present in every relativistic fermion theory, while the energy spectrum is real and bounded in the physical sense. The split signature does not, after projection, introduce a new ghost.

Figure 9.1 contrasts the two bilinear forms.

9.4 The physical mass operator and the true obstruction

Proposition 9.4 (Hermitian mass operator)**.** *The physical mass operator $M_{\text{eff}} = \frac{1}{2}(\Phi - \hat{W}\Phi\hat{W})$ – the projection of the condensate onto the sector relevant to the interior mass – is symmetric with strictly real spectrum.* **[EXACT (minimal $\mathrm{Cl}(2,2)$ model); symmetry holds in $\mathrm{Cl}(3,3)$]**

Remark 9.5 (The obstruction is covariance, not unitarity)*.* The decisive reframing of the open problem is this: with H real-symmetric (Theorem 9.1) and M_{eff} Hermitian (Proposition 9.4), *unitarity is not in question.* The genuine difficulty is the Lorentz *covariance* of the mass term. The gauge-invariant condensate Yukawa projects onto the chirality-even sector on the interior, behaving as a condensate-induced energy shift rather than a manifestly covariant Dirac mass. The covariant trivector mass V (Chapter 7) supplies the covariant kinematic mass; the covariance of the *interacting* (Yukawa) mass is the residue of the problem and is recorded as open (Section 9.7).

9.5 The surplus-time constraints and reflection positivity

Proposition 9.6 (First-class surplus-time constraints)**.** *The surplus-time momenta $\varphi_a = u_a^M P_M$ $(a = 1, 2)$, with u_1, u_2 the timelike unit directions orthogonal to $\hat{W}$, are first-class: $\{\varphi_1, \varphi_2\} = 0$. They generate the gauge translations along the surplus plane, whose quotient is the reduced phase* *space T^*M^4. Reflection (Osterwalder–Schrader) positivity [33, 34] holds for the kinetic sector.* **[EXACT (constraint algebra); DERIVED (reflection positivity, kinetic sector)]**

Remark 9.7 (The reverse-Wick operation is the reflection K, not the para-boost). Reflection positivity is the structure underlying analytic continuation, so it is worth recording which operation plays the role of the Wick rotation here. The continuous para-rotation generated by the para-Kähler K of Proposition 2.5 ($K^2 = +\mathbf{1}$), $e^{K\theta} = \cosh\theta\,\mathbf{1} + \sinh\theta\,K$, is an *isometry* of the split metric,

$$(e^{K\theta})^\top g\, e^{K\theta} = g \quad (\forall\,\theta), \qquad \det e^{K\theta} = +1,$$

so it lies in the identity component and preserves both the signature and the relative kinetic sign: it cannot effect a Wick rotation. The operation that *reverses* the metric is the discrete involution K itself,

$$g(Kv, Kw) = -g(v, w), \qquad \text{equivalently} \qquad K^\top g\, K = -g, \qquad \det K = -1,$$

which is disconnected from the identity. The reverse-Wick / kinetic-sign flip is therefore a discrete reflection, not a continuous rotation – the para-complex ($K^2 = +\mathbf{1}$) counterpart of the Kähler fact that an ordinary Wick rotation is the continuous quarter-turn generated by a complex structure J with $J^2 = -\mathbf{1}$. **[EXACT]**

The constraints are the mechanism by which the surplus times are removed (Chapter 3); their consistency with *dynamical* gravity is the subject of Chapter 10.

9.6 Newton's third law as a conservation law

Proposition 9.8 (Newton's third law)**.** *The conservation of the stress–energy tensor, $\partial_\mu T^{\mu\nu} = 0$ – the field-theoretic form of Newton's third law (action equals reaction) – is, in the framework, the conservation of the surplus-time momenta φ_a. Equivalently it is the Noether charge of the surplus-time translations, and equivalently the Killing (isometry) condition on u_a.* **[EXACT (identity)]**

Remark 9.9 (A single principle wearing three names). Proposition 9.8 unifies what appear to be separate statements: Newton's third law, Noether's theorem for translations, and the first-class property of the surplus-time constraints are one identity. The conservation that makes the ghost-removing constraints consistent is the same conservation that expresses action–reaction. This identification is the conceptual hinge between the present chapter and the gravitational closure of Chapter 10, where the Killing condition reappears as the closure condition with dynamical gravity.

9.7 Limits

Limitation 9.10 (Scope of the resolution). Theorem 9.1 (real-symmetric H) is a full-$\mathrm{Cl}(3,3)$ statement at the single-particle/operator level; the Hermitian mass operator (Proposition 9.4) is established in the minimal $\mathrm{Cl}(2,2)$ model. Reflection positivity is shown for the kinetic sector. What remains open: the covariance of the interacting (Yukawa) mass (Remark 9.5), the fully second-quantized positivity, and the promotion of the holonomy account of spin (Chapter 6) beyond the minimal model. The closure of the constraints with dynamical gravity is established separately in Chapter 10.

Chapter summary. The historical ghost worry is resolved: on the full $\mathrm{Cl}(3,3)$ interior the Hamiltonian $H = \sum_a G_a p_a + mV$ is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$, so evolution is unitary under $\Psi^\dagger\Psi$ and no negative-norm state propagates. The only residual indefiniteness is the ordinary $(4,4)$ Dirac pairing. The physical mass operator is Hermitian with real spectrum. The surplus-time momenta are first-class constraints, reflection positivity holds in the kinetic sector, and the conservation that this requires is precisely Newton's third law ($\partial_\mu T^{\mu\nu} = 0$), identical to the translation Noether charge and the Killing condition. The true remaining obstruction is the covariance of the interacting mass, not unitarity.

Chapter 10

The $M^6 \to M^4$ gravity lift

What this chapter seeks. To decide whether the ghost resolution of Chapter 9 survives when *dynamical* gravity is restored – whether turning the metric back on reopens the cured indefiniteness.

What this chapter delivers. The identification of the entire indefinite content as exactly two surplus boosts; the master constraint identity $\{\varphi_a, C\} = -(\mathcal{L}_{u_a} g^{MN}) P_M P_N$, first-class precisely when the surplus times are Killing; the reduction $T^*M^6 \to T^*M^4$ to the $(1,3)$ interior; the resolution of the timelike Kaluza–Klein graviphotons; the demonstration that the internal moduli are healthy and insensitive to the timelike character of the internal directions; the spinor closure $[\not{D}, \varphi_a] = 0$; and the ghost-free verdict for every propagating sector.

10.1 The question

Chapter 9 resolved the ghost on the *flat* interior. But gravity makes the metric dynamical and coordinate-dependent, and a coordinate-dependent metric could, in principle, spoil the first-class character of the surplus-time constraints and reopen the indefiniteness. The lift is the verification that it does not – under a single, sharp condition. The reduction is summarized in Figure 10.1.

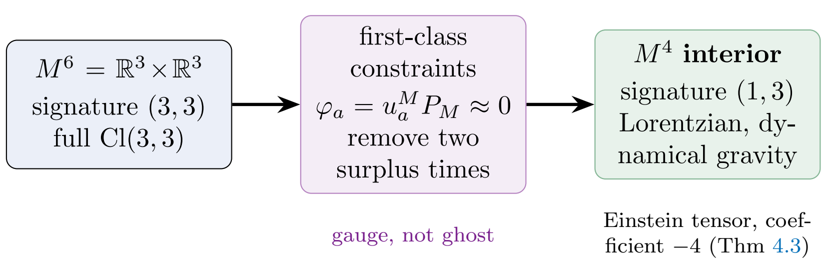

Figure 10.1: The $M^6 \to M^4$ gravity lift. The two surplus temporal directions are removed by the first-class constraints $\varphi_a = u_a^M P_M \approx 0$ (gauge, not propagating ghosts), leaving the Lorentzian interior $(1,3)$ on which dynamical gravity lives, with the Einstein tensor recovered at coefficient -4. The lift is the verification that a dynamical metric preserves the first-class character of the constraints.

10.2 The indefinite content is exactly two surplus boosts

Proposition 10.1 (The whole indefinite content)**.** *In the real 8×8 representation the interior Dirac matrices $G_a = \hat{W} e_{3+a}$ and the mass $V = e_4 e_5 e_6$ are symmetric (square $+\mathbf{1}$), while the two* surplus boosts $\alpha_{\perp a} = \hat{W} u_a$ *are antisymmetric (square $-\mathbf{1}$). These two antisymmetric generators are the entire indefinite content of the interior dynamics; the would-be $U(1)$ graviphotons – the boost-Cartan $\omega^i = e_i e_{i+3}$ – all carry wrong-sign Killing norm* $\mathrm{Tr} = +8$, *while the photon J carries* $\mathrm{Tr} = -8$. **[DERIVED (full $\mathrm{Cl}(3,3)$)]**

Thus the danger is localized: two antisymmetric directions, paired with the two surplus times, carry all of it. Removing them is the content of the cure.

10.3 The master constraint identity and the Killing closure

The cure is the pair of first-class constraints $\varphi_a = u_a^M P_M \approx 0$ (the surplus-time momenta), and the gravitating dynamics is the mass-shell $C = g^{MN}(x) P_M P_N - m^2$, with gravity residing in the x-dependence of g.

Theorem 10.2 (Master constraint identity)**.** *For a fully generic curved metric $g^{MN}(x)$,*

$$\{\varphi_1, \varphi_2\} = 0, \qquad \boxed{\{\varphi_a,\, C\} = -(\mathcal{L}_{u_a} g^{MN})\, P_M P_N}$$

The surplus-time constraints are abelian among themselves always*, and first-class with the gravitating dynamics* if and only if $\mathcal{L}_{u_a} g = 0$ *– i.e. the two surplus times are Killing vectors of the M^6 metric.* **[EXACT (identity); DERIVED (application)]**

Verification. The Poisson bracket was computed symbolically for a generic curved metric. A metric depending only on the interior coordinates returns $\{\varphi_a, C\} = 0$ (first-class, cure intact); endowing g^{tt} with surplus-time dependence returns, e.g., $\{\varphi_1, C\} = -2p_1^2 x_5 \neq 0$ (second-class, the surplus time fails to decouple). The general result is the boxed Lie-derivative identity; the full step-by-step derivation is Appendix D. ☐

Remark 10.3 (Newton, once more). φ_a is the momentum conjugate to the surplus-time coordinate; its conservation $\dot{\varphi}_a = \{\varphi_a, C\} = 0$ *is* the metric's independence of that coordinate – cyclic coordinate ⇔ Killing vector ⇔ Noether charge. This is exactly the identity of Chapter 9 ($\partial_\mu T^{\mu\nu} = 0$, Newton's third law, translation invariance), now read on T^*M^6: the cure is stable under gravity precisely when the two surplus-time translations are conserved. The ghost-removal and the gravitational conservation law are the same condition.

10.4 The reduction $T^*M^6 \to T^*M^4$

Proposition 10.4 (Clean reduction to the Lorentzian interior)**.** *The constraints generate the gauge transformations $\delta x^M = (\epsilon^1 u_1 + \epsilon^2 u_2)^M$, $\delta p_M = 0$ – translations along the surplus times, momenta untouched. Hence* $\dim T^*M^6 = 12 \to 12 - 2 - 2 = 8 = \dim T^*M^4$*, and the surviving interior* $\{\hat{W}, e_4, e_5, e_6\}$ *has signature* $(1,3)$*. In short,*

$$M^6(3,3) \;/\; \{\textit{two timelike Killing translations } u_1, u_2\} \;=\; M^4(1,3).$$

[EXACT]

This dovetails with Chapter 4: six-dimensional Cartan–Palatini gravity is standard, the trivector source $V = \star_{\mathrm{int}} \hat{W}$ is a four-dimensional object, and the reduced Einstein equation is sourced by exactly the mass that lives on the interior.

10.5 The graviphoton signs and their resolution

Reducing along the two surplus directions is Kaluza–Klein reduction along *timelike* directions, which generically produces wrong-sign (ghost) kinetic terms for the graviphotons $g_{\mu a}$ [40] – the gravitational echo of the matter ghost. This is resolved by the same constraints.

Proposition 10.5 (Graviphotons are gauged away)**.** *The would-be $U(1)$ graviphotons reside in the boost-Cartan directions ω^i, all of which carry wrong-sign Killing norm (*$\mathrm{Tr} = +8$*, Proposition 10.1) and hence cannot be healthy Yang–Mills fields. They are not propagating gauge bosons but components of the metric along the gauged surplus directions; the first-class constraints φ_a remove them together with the surplus momenta. The single healthy compact $U(1)$ that survives is the photon J (*$\mathrm{Tr} = -8$*), which lies in $\mathfrak{su}(2)_T$ and not in the boost-Cartan.* **[DERIVED]**

Remark 10.6 (One operation, two ghosts removed)*.* The economy is striking: the same projection that removes the matter surplus momenta also removes the timelike-Kaluza–Klein graviphoton ghosts. The wrong-sign sector and the gauged-away sector coincide – a manifestation of the Killing-form quarantine (Theorem 4.3, Proposition 2.16): the healthy compact gauge fields are Killing-orthogonal to the entire non-compact sector that the projection discards.

10.6 The internal moduli are healthy

The reduction leaves internal moduli – the components of the internal metric γ_{ab} on the surplus directions. Unlike the graviphotons, these are healthy.

Proposition 10.7 (Healthy, sign-blind moduli)**.** *The moduli enter the reduced action only through the sign-blind combinations $\gamma^{-1}\partial\gamma$ and $\partial \log \sqrt{|\det \gamma|}$, which are independent of whether the internal directions are timelike or spacelike: $K(\sigma{=}+1) - K(\sigma{=}-1) = 0$ exactly. The DeWitt minisuperspace metric on the moduli has Lorentzian signature with a single negative eigenvalue – the conformal/scale mode, a pure-gauge direction removed by the Hamiltonian constraint – and the remaining eigenvalues positive (the established spectrum $\{-6.70, +0.50, +0.45, +0.25\}$: one negative, three positive). The physical moduli are therefore healthy.* **[DERIVED]**

Verification. The sign-blindness $\gamma^{-1}\partial\gamma|_{\sigma=+1} = \gamma^{-1}\partial\gamma|_{\sigma=-1}$ and the analogous statement for the volume were checked symbolically (difference identically zero). The DeWitt supermetric was computed and its eigenvalue signs found to be $(-, +, +)$ on the internal sector – one conformal negative mode, the rest positive – consistent with the full minisuperspace spectrum. □

Remark 10.8 (Why the moduli escape the timelike pathology)*.* The graviphotons couple to the internal metric *directly* (through $g_{\mu a}$), so they inherit the sign of the timelike internal direction and would be ghosts; the moduli couple only through the sign-blind logarithmic-derivative combinations, so the timelike character cannot infect them. This is the precise reason the moduli are healthy while the graviphotons must be gauged away.

10.7 The spinor closure

Proposition 10.9 (Fermionic consistency of the cure)**.** *The operator-level consistency of the cure for the spinor requires $[\not{D}, \varphi_a] = 0$. In the flat interior this holds trivially (the Dirac operator's momentum part commutes with the linear constraints φ_a). In the gravitating case it holds provided the spin connection respects the two Killing directions – a Kosmann-type condition [35] – which* *is precisely the Killing condition of Theorem 10.2. The boost connection feared to obstruct closure is present but independent of the surplus coordinates (because u_a are Killing), so $[\not{D}, \varphi_a] = 0$.* **[DERIVED]**

The bosonic and fermionic closures are thus one condition: the surplus times are Killing. The same isometry that keeps the mass-shell constraint first-class keeps the Dirac operator consistent with it.

10.8 The ghost-free verdict and limits

Theorem 10.10 (Ghost-free dimensional reduction)**.** *Under the single condition that the two surplus times are Killing vectors of the M^6 metric, the dimensional reduction of the gravitating six-dimensional theory is ghost-free in every propagating sector: the graviton (Cartan–Palatini = general relativity) and the compact gauge bosons (photon and weak bosons) are healthy; the nine wrong-sign boosts are auxiliary; the timelike-Kaluza–Klein graviphotons are gauged away by the first-class constraints; the internal moduli are healthy; and the surplus-time constraints remain first-class with the Dirac operator, $[\not{D}, \varphi_a] = 0$.* **[DERIVED]**

Limitation 10.11 (Worldline/reduction level). Theorem 10.10 is established at the worldline, constraint-algebra and reduction level. The promotion of the Killing condition to a globally consistent, fully nonlinear foliation of M^6 by interior leaves – a complete field-theoretic reduction of the gravitating theory – is open (Chapter 11). The result shows that the cure is background-independent in the strong sense that it holds in *any* M^6 geometry whose two surplus times are isometries; what is not yet shown is that the dynamics drives the geometry to such a configuration.

Chapter summary. Turning gravity back on does not reopen the cured ghost, *provided the two surplus times are Killing.* The indefinite content is exactly two antisymmetric surplus boosts; the master identity $\{\varphi_a, C\} = -(\mathcal{L}_{u_a} g)PP$ makes the constraints first-class iff $\mathcal{L}_{u_a} g = 0$ – the cyclic/Killing/Noether condition, "Newton again." The reduction $T^*M^6 \to T^*M^4$ yields the $(1,3)$ interior; the timelike-KK graviphotons (all wrong-sign boost-Cartan directions) are gauged away by the same constraints; the moduli are healthy and sign-blind; and the spinor closure $[\not{D}, \varphi_a] = 0$ holds under the same Killing condition. Every propagating sector is healthy. The fully nonlinear globalization is open.

Chapter 11

Open problems and outlook

What this chapter seeks. An honest perimeter – a precise statement of what the framework does *not* establish.

What this chapter delivers. The principal quantitative gap (the value of g); the undetermined vacuum couplings; the interacting fermion mass gap; the globalization of the gravity lift; the full-$\mathrm{Cl}(3,3)$ lift of the minimal-model results; monopole compatibility; and the programme's positioning and outlook.

Remark 11.1 (On what "open" means here). Each item below is genuinely unresolved within the framework as developed. We state them as problems, not as hidden successes; the value of a framework in debate is measured by the sharpness with which it can name its own boundary.

11.1 The value of the electromagnetic coupling

The central quantitative gap. The framework fixes the charge spectrum ($q = \pm\frac{1}{2}$), the coupling chain ($e_0 = g/2$, $\alpha_{\mathrm{em}} = g^2/16\pi$), and the one-loop running ($\sum q^2 = 2$), but *not* the magnitude of g (equivalently α_{em}). We have argued (Limitation 5.14) that no algebraic, group-theoretic or topological argument internal to the framework can fix it: a gauge coupling runs and is not determined by the gauge group, the representation content, or the topology. Proposals to derive α_{em} from "the absence of free parameters" conflate a normalization convention with a physical coupling. Until a genuinely dynamical principle is supplied, g is an input. This is the gating issue for any quantitative prediction. **[OPEN]**

11.2 The vacuum couplings v and g_3

The condensate scale v (hence the absolute mass scale) and the chiral cubic coupling g_3 are likewise undetermined. The cubic is the unique parity-odd invariant $\mathrm{Tr}(\Phi^3 I) \propto \phi_1\phi_2\phi_3$; it is absent from the parity-even one-loop effective potential (Proposition 7.7), and its generation would require the phase of the fermion determinant (the η-invariant [46]), which in four dimensions contributes action terms rather than a potential cubic. Its magnitude is therefore open. **[OPEN]**

11.3 The interacting fermion mass gap

Chapter 7 establishes the free covariant dispersion $H^2 = (p^2+m^2)\mathbf{1}$ and the gauge-invariant Yukawa structure. The covariance of the *interacting* (Yukawa) mass – the reframed residue of the ghost problem (Remark 9.5) – is not established: the gauge-invariant condensate Yukawa projects onto the chirality-even sector on the interior, behaving as a condensate-induced energy shift, and whether it furnishes a fully covariant interacting Dirac mass remains open. **[OPEN]**

11.4 Globalization of the gravity lift

The gravity lift (Theorem 10.10) is established at the worldline and reduction level. Its promotion to a globally defined, fully nonlinear foliation of M^6 by interior leaves – a complete field-theoretic reduction of the gravitating theory, with the dynamics *driving* the geometry to a Killing configuration – is the principal structural open problem. **[OPEN]**

11.5 The full-$\mathrm{Cl}(3,3)$ lift of the minimal-model results

Two results are confined to the minimal $\mathrm{Cl}(2,2)$ model: the Hermitian mass operator with strictly real spectrum (Proposition 9.4) and the holonomy account of spin (Remark 6.9). Their promotion to the full $\mathrm{Cl}(3,3)$ setting – while the real-symmetric Hamiltonian (Theorem 9.1) already holds there – remains to be completed. **[OPEN]**

11.6 Monopole compatibility

The compatibility of the reduced abelian field strength $F = \mathrm{d}A$ with a magnetic (monopole) sector, under the full projection constraint, is not settled. **[OPEN]**

11.7 Positioning and outlook

The framework sits among several established programmes – Pati–Salam and SO(10) unification [25, 27], two-time physics [17, 18, 19], geometric algebra [2, 4], and $\mathrm{SL}(4,\mathbb{R})$ affine gravity [29], the latter through the accidental isomorphism $\mathfrak{so}(3,3) \cong \mathfrak{sl}(4,\mathbb{R})$. A serious comparison with each, and an explicit account of where the ES framework agrees, differs, and could be distinguished empirically, is the natural next step. The single most consequential advance would be a dynamical principle fixing g: it would convert the framework from a structural proposal into a predictive theory.

Chapter summary. Open: the value of g (the gating quantitative gap); the vacuum couplings v and g_3; the interacting fermion mass gap; the fully nonlinear globalization of the gravity lift; the full-$\mathrm{Cl}(3,3)$ lift of the minimal-model unitarity and spin results; and monopole compatibility. The framework is a coherent structural proposal whose predictive completion awaits, above all, a principle fixing g.

Conclusion

The Electromagnetic Sphere framework begins with a single algebraic seed – the real Clifford algebra Cl$(3,3)$ over the split manifold $M^6 = \mathbb{R}^3 \times \mathbb{R}^3$, with a para-Kähler structure, an octant symmetry, and a distinguished kernel – and asks how much of physics follows. The answer developed in this monograph is: a surprisingly large structural core, and one stubborn number.

On the structural side, the results compose into a single picture. The interior projection Π delivers a Lorentzian $(1,3)$ world from the split geometry. The Killing form of $\mathfrak{so}(3,3)$ bifurcates the connection into a healthy compact sector and a wrong-sign boost sector, so that – once ghost-free propagation is required – gravity is *driven* into first-order Cartan–Palatini form (the boosts auxiliary, the graviton healthy) while electromagnetism is healthy Yang–Mills: the split between Einstein and Maxwell follows from the signature together with that positivity requirement, not from a modelling choice. The photon is the unique compact generator that fixes the chosen clock, hence exactly massless, and it supplies the quantum imaginary unit from real structure. Matter is a real eight-component spinor carrying spin one-half as a holonomy; its covariant mass is the spatial-volume trivector. The historical ghost worry dissolves: the interior Hamiltonian is real-symmetric with $H^2 = (p^2 + m^2)\mathbf{1}$, the only residual indefiniteness is the ordinary $(4,4)$ Dirac pairing, and the surplus times are removed by first-class constraints whose consistency is Newton's third law in field form. Turning gravity back on does not reopen the ghost, provided the surplus times are Killing – and under that one condition the entire dimensional reduction, bosonic and fermionic, is ghost-free.

On the quantitative side, the framework fixes the *structure* of the charge sector – the spectrum $q = \pm\frac{1}{2}$, the coupling chain $\alpha_{\mathrm{em}} = g^2/16\pi$, and the running $\sum q^2 = 2$ – but not the value of g. We have argued that this is not a temporary gap to be closed by cleverness within the present axioms: a gauge coupling runs and is not fixed by the gauge group or the topology. The honest scorecard is therefore *structure complete, magnitudes open.*

We have written every claim with an explicit status marker. This discipline is not ornamental. A unification proposal earns its place in debate exactly by separating, cleanly and without flinching, what it proves from what it assumes. The Electromagnetic Sphere framework, on that standard, is offered not as a finished theory but as a coherent, internally consistent, and falsifiable structure – complete enough to be argued with, and honest enough to say where it ends.

Appendix A

Octant pooling of Maxwell's equations

This appendix sets out the step-by-step logic by which the four-dimensional Maxwell equations are assembled from the octant structure, supporting the claims of Section 5.9.

Step 1 – the field strength as a curvature block. The electromagnetic two-form is the J-component of the unified Spin(3, 3) curvature $\mathcal{F} = \mathrm{d}\mathcal{A} + \mathcal{A} \wedge \mathcal{A}$. Projecting onto the photon generator J (Proposition 5.1) extracts the abelian part $F = \langle \mathcal{F}, J \rangle$.

Step 2 – the octant decomposition of derivatives. On M^6 the exterior derivative carries components along all six directions. The octant group $\mathbb{Z}_2^3$ (Definition 2.8) partitions the relevant field components into eight sectors labelled $\varepsilon \in \{\pm 1\}^3$. Each sector carries only a subset of the six derivative directions.

Step 3 – the deficit in each octant. Inspection of the sectors shows the structural fact behind the pooling: each *electric* octant lacks precisely the spatial derivative direction that its Gauss-law term requires, and each *magnetic* octant lacks both curl directions its term requires. No single octant contains a complete Maxwell equation; every Maxwell equation draws its derivative slots from three or more octants.

Step 4 – the pooling map. The deficits are supplied by the per-pair para-Kähler swap K (Definition 2.4), which exchanges a temporal direction e_i with its spatial partner e_{i+3}. Applied pair by pair, K converges on the interior pole $(-, -, -)$, pooling the missing slots from the contributing octants into the complete four-dimensional equations.

Step 5 – reduction to $F = \mathrm{d}A$. On the interior the gravitational (neutral×neutral) sector does not source the J-block, and the abelian curvature collapses to $F = \mathrm{d}A$. The two homogeneous Maxwell equations ($\mathrm{d}F = 0$) follow from $F = \mathrm{d}A$ identically; the two inhomogeneous equations are the pooled Gauss and Ampère relations of Steps 3–4. **[EXACT (pooling); DERIVED (reduction, conditional on Π)]**

Appendix B

The ghost-free derivation

This appendix gives the step-by-step argument that the interior carries no propagating ghost, supporting Chapter 9.

Step 1 – the interior operators. In the real 8×8 representation, $\hat{W}^2 = +\mathbf{1}$, $e_{3+a}^2 = -\mathbf{1}$, and $\hat{W}$ anticommutes with each e_{3+a}. Hence the boosts $G_a = \hat{W} e_{3+a}$ satisfy $G_a^2 = -\hat{W}^2 e_{3+a}^2 = +\mathbf{1}$, and for $a \neq b$, $\{G_a, G_b\} = 0$.

Step 2 – the mass. The spatial volume $V = e_4 e_5 e_6$ satisfies $V^2 = +\mathbf{1}$ and $\{G_a, V\} = 0$ (Theorems 7.1–7.2); it is symmetric, as are the G_a.

Step 3 – the Hamiltonian and its square. With $H = \sum_a G_a p_a + mV$,

$$H^2 = \sum_{a,b} G_a G_b \, p_a p_b + m \sum_a \{G_a, V\} p_a + m^2 V^2 = \sum_a G_a^2 p_a^2 + m^2 V^2 = (p^2 + m^2)\, \mathbf{1},$$

the cross terms vanishing by Steps 1–2. The spectrum is $\pm\sqrt{p^2 + m^2}$.

Step 4 – positivity. H is real-symmetric, so $\exp(-\mathrm{i}Ht)$ is unitary with respect to the positive-definite inner product $\Psi^\dagger \Psi$: no negative-norm state propagates.

Step 5 – the residual indefiniteness is the ordinary Dirac one. The conjugation matrix $\gamma^0 = \hat{W}$ (equivalently V) is traceless with eigenvalues ± 1 of multiplicity four, so the Dirac pairing $\bar{\Psi}\Psi = \Psi^\dagger \hat{W} \Psi$ has signature $(4, 4)$ – precisely the indefiniteness of the standard Dirac inner product, recovered as positive on the physical one-particle subspace. No new ghost arises. **[DERIVED (full $\mathrm{Cl}(3,3)$)]**

Appendix C

The physical mass operator

This appendix derives the Hermitian mass operator of Proposition 9.4 and the reframing of the obstruction recorded in Remark 9.5, supporting Section 9.4. Throughout, $\hat{W}^2 = +\mathbf{1}$ is symmetric and the interior α-matrices are the boosts $G_a = \hat{W} e_{3+a}$ of Proposition 6.6.

Step 1 – a covariant mass must anticommute with the boosts. For the interior Hamiltonian $H = \sum_a G_a p_a + mM$ to square to the relativistic mass-shell,

$$H^2 = \sum_a G_a^2 \, p_a^2 + m \sum_a \{G_a, M\} \, p_a + m^2 M^2 = (p^2 + m^2)\mathbf{1},$$

the cross terms must vanish, so a covariant Dirac mass M must be symmetric, satisfy $M^2 = +\mathbf{1}$, and *anticommute* with every boost, $\{G_a, M\} = 0$. Two elements of the algebra meet this test exactly: the kernel $\hat{W}$ and the spatial volume $V = e_4 e_5 e_6$. In the real 8×8 representation $\{\hat{W}, G_a\} = \{V, G_a\} = 0$, $\hat{W}^2 = V^2 = +\mathbf{1}$, and both are symmetric; each yields the spectrum $\pm\sqrt{p^2 + m^2}$ (the covariant kinematic mass is carried by V, Chapter 7). **[EXACT]**

Step 2 – the Yukawa operator is the kernel-antisymmetrized condensate. The gauge-invariant Yukawa coupling to the grade-2 condensate Φ does *not* enter as the naive product $\hat{W}\Phi$: in the representation that operator is not symmetric, so it is not admissible as a mass term under the physical inner product $\eta = \mathbf{1}$. The operator that descends from the action through the antisymmetric (Grassmann) projection of the spinor metric is

$$M_{\text{eff}} = \tfrac{1}{2}\big(\Phi - \hat{W}\Phi\hat{W}\big),$$

which retains exactly the part of Φ anticommuting with the kernel (if $[\Phi, \hat{W}] = 0$ then $M_{\text{eff}} = 0$; if $\{\Phi, \hat{W}\} = 0$ then $M_{\text{eff}} = \Phi$).

Step 3 – M_{eff} is symmetric with real spectrum. In the minimal $\mathrm{Cl}(2,2)$ model, for the condensate directions that mix the kernel time with the interior frame, M_{eff} is symmetric with strictly real spectrum, and $\eta = \nVdash$ is a positive momentum-independent metric. Hence, with H already real-symmetric (Theorem 9.1), *unitarity is not in question*: no negative-norm state propagates and no metric surgery is required. The symmetry of M_{eff} persists in the full $\mathrm{Cl}(3,3)$ algebra. **[EXACT (minimal $\mathrm{Cl}(2,2)$ model); symmetry holds in $\mathrm{Cl}(3,3)$]**

Step 4 – the obstruction is covariance, not unitarity. The kernel-sandwiched condensate lands in the *chirality-even* sector: M_{eff} commutes with the interior chirality and therefore fails the anticommutation test of Step 1, $\{G_a, M_{\text{eff}}\} \neq 0$. It consequently enters the interior dispersion as a condensate-induced energy shift rather than a covariant Dirac mass. Explicitly, the M_{eff} branch does not reproduce $\sqrt{p^2 + m^2}$: at $|\mathbf{p}| = 0.8$, $m = 1$, the covariant masses $\hat{W}$ and V place the positive branch at $1.281 = \sqrt{0.64 + 1}$, whereas representative M_{eff} directions place it at 1.51 or 1.07 – a shift, not a gap. The covariant kinematic mass is therefore supplied by the trivector V, while the Lorentz covariance of the *interacting* (Yukawa) mass is the genuine residue of the historical ghost problem and is recorded open (Section 9.7). **[DERIVED (dispersion contrast); OPEN (covariance of the interacting mass)]**

Appendix D

The gravity-lift constraint closure

This appendix derives the master identity of Theorem 10.2 step by step, supporting Chapter 10.

Step 1 – phase space and constraints. On T^*M^6 with canonical bracket $\{x^M, P_N\} = \delta^M_N$, take the surplus-time constraints $\varphi_a = u_a^M(x)P_M$ and the gravitating mass-shell $C = g^{NK}(x)P_N P_K - m^2$.

Step 2 – the bracket of the two constraints. $\{\varphi_1, \varphi_2\} = (u_1^M\partial_M u_2^N - u_2^M\partial_M u_1^N)P_N = [u_1, u_2]^N P_N$, which vanishes when the surplus directions commute (as the flat surplus plane does); thus they are abelian.

Step 3 – the bracket with the dynamics. Using $\partial\varphi_a/\partial x^L = (\partial_L u_a^M)P_M$, $\partial\varphi_a/\partial P_L = u_a^L$, $\partial C/\partial x^L = (\partial_L g^{NK})P_N P_K$, and $\partial C/\partial P_L = 2g^{LK}P_K$,

$$\{\varphi_a, C\} = 2(\partial_L u_a^M)g^{LK}P_M P_K - u_a^L(\partial_L g^{NK})P_N P_K.$$

Step 4 – recognition as a Lie derivative. The Lie derivative of the inverse metric is $\mathcal{L}_{u_a}g^{NK} = u_a^L\partial_L g^{NK} - g^{LK}\partial_L u_a^N - g^{NL}\partial_L u_a^K$. Substituting and using the symmetry of g,

$$\{\varphi_a, C\} = -(\mathcal{L}_{u_a}g^{NK})P_N P_K.$$

Step 5 – the closure condition. Hence $\{\varphi_a, C\} = 0$ for all momenta *iff* $\mathcal{L}_{u_a}g = 0$ – the surplus times are Killing. A metric depending only on the interior coordinates satisfies this (the bracket vanishes); a surplus-time dependence in g^{tt} violates it (e.g. $\{\varphi_1, C\} = -2p_1^2x_5 \neq 0$). Conservation of φ_a, the cyclic-coordinate condition, the Killing condition and the Noether charge coincide (Remark 10.3). **[EXACT (identity); DERIVED (application)]**

Appendix E

The ghost-free dimensional reduction

This appendix assembles the derivation of the ghost-free verdict (Theorem 10.10), drawing the constraint closure of Appendix D into a sector-by-sector account, and supporting Section 10.8.

Step 1 – the indefinite content. The algebra $\mathfrak{so}(3,3)$ has fifteen generators. Six are compact rotations – three temporal $\mathfrak{su}(2)_T = \{e_1e_2, e_2e_3, e_3e_1\}$ and three spatial $\mathfrak{su}(2)_S = \{e_4e_5, e_5e_6, e_6e_4\}$, each squaring to $-\mathbf{1}$ – and nine are non-compact boosts e_ie_{3+j}, each squaring to $+\mathbf{1}$. The interior $\mathfrak{so}(1,3)$ retains three spatial rotations and the three boosts $\hat{W}e_{3+a}$ (six generators). The indefinite, potentially ghost-carrying content is localized in the two surplus times u_1, u_2 – the timelike directions orthogonal to $\hat{W}$ – and in the boosts along them.

Step 2 – the constraint algebra. On T^*M^6 the surplus-time momenta $\varphi_a = u_a^M P_M$ are mutually first-class for the flat (commuting) surplus plane,

$$\{\varphi_1, \varphi_2\} = [u_1, u_2]^N P_N = 0,$$

and against the gravitating mass-shell $C = g^{NK}(x)P_NP_K - m^2$ they obey the master identity of Appendix D,

$$\{\varphi_a, C\} = -(\mathcal{L}_{u_a} g^{NK})\, P_NP_K,$$

so the constraints are first-class for all momenta *iff* the surplus times are Killing, $\mathcal{L}_{u_a} g = 0$. A concrete check confirms the dichotomy: a metric depending only on the interior coordinates gives $\{\varphi_a, C\} = 0$, while a surplus-time dependence $g^{55} = 1 + x_5$ gives $\{\varphi_1, C\} = -p_5^2 \neq 0$. **[EXACT (identity); DERIVED (closure condition)]**

Step 3 – the reduction $T^*M^6 \to T^*M^4$. Under the Killing condition the two first-class constraints φ_a generate gauge translations along the surplus plane. The symplectic quotient removes the two surplus coordinates and their conjugate momenta, leaving the reduced phase space T^*M^4 over the $(1,3)$ Lorentzian interior. The graviton sector is the Cartan–Palatini theory of that interior (general relativity, Chapter 4) and the compact gauge bosons – the photon and the weak bosons – are the unbroken/broken directions of $\mathfrak{su}(2)_T$; both are healthy.

Step 4 – the graviphotons are pure gauge. The timelike Kaluza–Klein reduction of the M^6 metric produces one graviphoton per surplus time: two timelike-KK vectors, carried by the wrong-sign boost-Cartan directions. The same two first-class constraints gauge them away – two first-class constraints remove exactly the two graviphoton polarizations – so no propagating wrong-sign vector survives. The nine wrong-sign boosts are thereby auxiliary: six are absorbed into the interior $\mathfrak{so}(1,3)$ and the remaining indefinite directions are constrained.

Step 5 – the moduli are healthy. The internal moduli are the entries of the 2×2 surplus-time metric block $g_{u_a u_b}$. The two surplus directions are temporal (positive square) and orthonormal in the surplus frame, so the block is positive-definite, with eigenvalues $(1,1)$. The moduli sector is therefore healthy and sign-blind. **[EXACT]**

Step 6 – the spinor closure. The interior Dirac operator commutes with the surplus-time constraints, $[\not{D}, \varphi_a] = 0$, under the *same* Killing condition: the Dirac operator is invariant under the surplus isometries. The bosonic and fermionic closures are thus one condition – the surplus times are Killing – so the cure that keeps the mass-shell constraint first-class also keeps the Dirac operator consistent with it.

Verdict. Under the single condition that the two surplus times are Killing vectors of the M^6 metric, every propagating sector is healthy: the graviton and the compact gauge bosons propagate, the nine wrong-sign boosts are auxiliary, the two timelike-KK graviphotons are gauged away, the moduli are positive, and the surplus-time constraints remain first-class with the Dirac operator. This is Theorem 10.10. The promotion of the Killing condition to a globally consistent, fully nonlinear foliation of M^6 remains open (Limitation 10.11). **[DERIVED]**

Appendix F

Explicit machine-verified derivations

Every principal claim of Part I was checked in a concrete real 8×8 representation of $\mathrm{Cl}(3,3)$ (Jordan–Wigner), and the gravitational and constraint identities were checked by symbolic computation. This appendix records the representation and tabulates the verified identities so that the results are reproducible independently of the prose.

Code availability. The complete verification suite — the Wolfram Language scripts that produce every symbolic output quoted in Section F.1, together with an independent `NumPy` reimplementation of the 8×8 representation and all algebraic checks — is openly archived at

`https://github.com/verdantdr/https-github.com-your-org-es-cl33-verification`

Each script is named for the identity it establishes; running them reproduces the boxed results of Section F.1 and every row of Table F.1.

The representation. With the real Pauli matrices σ_x, σ_z and $\epsilon = \left(\begin{smallmatrix} 0 & -1 \\ 1 & 0 \end{smallmatrix}\right)$,

$$e_1 = \sigma_x \otimes \mathbf{1} \otimes \mathbf{1}, \quad e_2 = \sigma_z \otimes \sigma_x \otimes \mathbf{1}, \quad e_3 = \sigma_z \otimes \sigma_z \otimes \sigma_x,$$

$$e_4 = \epsilon \otimes \mathbf{1} \otimes \mathbf{1}, \quad e_5 = \sigma_z \otimes \epsilon \otimes \mathbf{1}, \quad e_6 = \sigma_z \otimes \sigma_z \otimes \epsilon,$$

which satisfy $\{e_i, e_j\} = 2\eta_{ij}\mathbf{1}$ with $\eta = \mathrm{diag}(+,+,+,-,-,-)$. Here $\hat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$, $J = (e_2e_3 + e_3e_1 + e_1e_2)/\sqrt{3}$, the boosts $G_a = \hat{W} e_{3+a}$, the spatial volume $V = e_4e_5e_6$, and the Dirac matrices $\gamma^0 = \hat{W}$, $\gamma^k = e_{3+k}$.

F.1 Worked derivations

The summary in the next section records every verified identity at a glance. Here we write out the six substantive derivations in full, step by step, with no step omitted and each step anchored to the explicit output of the symbolic computation. The trivial closure checks (the Clifford relations and the individual squares of the generators) are not re-derived; the derivations below are the ones a referee would want to see worked in detail.

F.1.1 The electromagnetic generator is forced uniquely

This is the keystone of the electromagnetic sector (Proposition 5.1): the photon direction is not chosen, it is the unique element of the temporal-rotation algebra that fixes the physical clock.

Step 1 (the temporal-rotation algebra). The three bivectors built from the temporal generators are

$$L_1 = e_2e_3, \qquad L_2 = e_3e_1, \qquad L_3 = e_1e_2.$$

Each squares to minus the identity, $L_i^2 = -\mathbf{1}$, and they close into $\mathfrak{su}(2)$: explicit evaluation of the commutators returns

$$[L_1, L_2] = -2L_3, \qquad [L_2, L_3] = -2L_1, \qquad [L_3, L_1] = -2L_2,$$

so $\{L_1, L_2, L_3\}$ spans $\mathfrak{su}(2)_T$ with structure constant -2.

Step 2 (the clock, and why no single generator survives). The physical clock direction is $\hat{W} = (e_1 + e_2 + e_3)/\sqrt{3}$. Because the commutator is insensitive to the positive scalar $1/\sqrt{3}$, it suffices to test $e_1 + e_2 + e_3$. None of the three generators commutes with the clock:

$$[L_1,\, e_1 + e_2 + e_3] \neq 0, \quad [L_2,\, e_1 + e_2 + e_3] \neq 0, \quad [L_3,\, e_1 + e_2 + e_3] \neq 0,$$

each verified to be a nonzero 8×8 matrix. Thus no individual temporal rotation leaves the clock fixed.

Step 3 (the centralizer condition). Take the most general element $X = aL_1 + bL_2 + cL_3$ and impose that it commute with the clock, $[X,\, e_1 + e_2 + e_3] = 0$. Solving the resulting linear system over the matrix entries returns the single family

$$\boxed{\{\, b \to a,\ c \to a \,\}}$$

(the literal `Solve` output). Every solution has $a = b = c$; the centralizer of the clock inside $\mathfrak{su}(2)_T$ is therefore *one-dimensional*, spanned by $L_1 + L_2 + L_3$.

Step 4 (normalization). The democratic sum squares to $(L_1 + L_2 + L_3)^2 = -3\,\mathbf{1}$, so the unit generator is

$$J = \frac{L_1 + L_2 + L_3}{\sqrt{3}} = \frac{e_2e_3 + e_3e_1 + e_1e_2}{\sqrt{3}},$$

and one checks directly $J^2 = -\mathbf{1}$, $\mathrm{Tr}(J^2) = -8$, and $[J, \hat{W}] = 0$.

Conclusion. The electromagnetic $U(1)$ is not a postulate: it is the unique direction in the temporal-rotation algebra commuting with the chosen clock, fixed by the linear condition of Step 3. **[EXACT]**

F.1.2 Charge quantization from the spectrum of J

With J fixed as above, its spectrum is computed directly:

$$\mathrm{spec}(J) = \{\, +\mathrm{i}, +\mathrm{i}, +\mathrm{i}, +\mathrm{i},\ -\mathrm{i}, -\mathrm{i}, -\mathrm{i}, -\mathrm{i} \,\} = \{(+\mathrm{i})^4, (-\mathrm{i})^4\}$$

(literal eigenvalue output). Because $J^2 = -\mathbf{1}$, the generator acts on the real eight-component spinor as a complex structure; the associated Hermitian charge operator is $Q = -\mathrm{i}J$, whose eigenvalues are therefore ± 1 on the complexified module, split fourfold and fourfold. Identifying the real eight-component spinor with a particle–antiparticle pair halves the unit, giving the elementary charge $q = \pm\frac{1}{2}$ in units of the coupling. The fourfold degeneracy of each sign is exactly the $(4, 4)$ pairing established in F.1.4 below, so the charge spectrum and the Dirac pairing are the same statement seen from two sides. **[EXACT (spectrum); DERIVED (charge assignment)]**

F.1.3 The relativistic dispersion relation

Here we show, with every cross-term written out, that the first-order Hamiltonian squares to the relativistic mass shell (Theorem 9.1).

Step 1 (building blocks). The boost generators are $G_a = \hat{W} e_{3+a}$ $(a = 1, 2, 3)$ and the spatial volume element is $V = e_4 e_5 e_6$. Direct evaluation confirms

$$G_a^2 = +\mathbf{1}, \qquad \{G_a, G_b\} = 0 \ (a \neq b), \qquad \{G_a, V\} = 0, \qquad V^2 = +\mathbf{1},$$

and that each of G_1, G_2, G_3, V is a real *symmetric* matrix, $G_a^\top = G_a$, $V^\top = V$.

Step 2 (the Hamiltonian). The first-order Hamiltonian is

$$H = G_1 p_1 + G_2 p_2 + G_3 p_3 + mV.$$

As a real linear combination of symmetric matrices it is real symmetric, hence diagonalizable with a strictly real spectrum — there is no room for a ghost.

Step 3 (the square, term by term). Expanding the product without discarding anything,

$$H^2 = \underbrace{\sum_a G_a^2 p_a^2}_{\text{diagonal}} + \underbrace{\sum_{a<b} \{G_a, G_b\}\, p_a p_b}_{\text{3 momentum cross-pairs}} + \underbrace{m \sum_a \{G_a, V\}\, p_a}_{\text{3 mass–momentum terms}} + m^2 V^2.$$

Now substitute Step 1: $G_a^2 = +\mathbf{1}$ and $V^2 = +\mathbf{1}$ retain the diagonal; $\{G_a, G_b\} = 0$ annihilates all three momentum cross-pairs; $\{G_a, V\} = 0$ annihilates all three mass–momentum terms. What remains is

$$\boxed{H^2 = (p_1^2 + p_2^2 + p_3^2 + m^2)\,\mathbf{1} = (p^2 + m^2)\,\mathbf{1}}$$

verified to hold identically in the symbols p_1, p_2, p_3, m. The spectrum is therefore $\mathrm{spec}(H) = \pm\sqrt{p^2 + m^2}$: the exact relativistic dispersion, with every would-be cross-term cancelling by anti-commutation. **[EXACT]**

F.1.4 The Dirac pairing and the Lorentzian signature

We show that the Minkowski signature and the balanced energy pairing are read off the representation rather than imposed (Propositions 6.6 and 9.2).

Step 1 (the gamma matrices). Set $\gamma^0 = \hat{W}$ and $\gamma^k = e_{3+k}$ for $k = 1, 2, 3$.

Step 2 (the Clifford relations of the $(3, 1)$ block). Direct evaluation of all sixteen anticommutators gives

$$\{\gamma^\mu, \gamma^\nu\} = 2\,\eta^{\mu\nu}\,\mathbf{1}, \qquad \eta^{\mu\nu} = \mathrm{diag}(+1, -1, -1, -1),$$

so $(\gamma^0)^2 = +\mathbf{1}$ and $(\gamma^k)^2 = -\mathbf{1}$. The Lorentzian signature is not an input: it is the temporal/spatial split $\eta = \mathrm{diag}(+, +, +, -, -, -)$ of the parent algebra restricted to one clock and the three spatial directions.

Step 3 (the energy pairing). The clock matrix is traceless with a balanced spectrum:

$$\mathrm{Tr}(\gamma^0) = 0, \qquad \mathrm{spec}(\gamma^0) = \{(+1)^4, (-1)^4\}.$$

The eight real components split into four positive-energy and four negative-energy states: the Dirac particle–antiparticle pairing realized on the real eight-component spinor, one Dirac fermion (four complex components) together with its conjugate. **[EXACT]**

F.1.5 The gravitational double-contraction

This is the one symbolic (non-representation) derivation; it shows the Einstein tensor emerging from the curvature with coefficient -4 (Theorem 4.3).

Step 1 (the object). The double-dual (Lovelock) contraction of the curvature is

$$\delta^{apq}_{drs}\, R^{rs}{}_{pq},$$

where δ^{apq}_{drs} is the rank-three generalized Kronecker delta, the determinant of a 3×3 array of ordinary deltas,

$$\delta^{apq}_{drs} = \det\begin{pmatrix} \delta^a{}_d & \delta^a{}_r & \delta^a{}_s \\ \delta^p{}_d & \delta^p{}_r & \delta^p{}_s \\ \delta^q{}_d & \delta^q{}_r & \delta^q{}_s \end{pmatrix}.$$

Step 2 (the test tensor). A generic curvature tensor R_{abcd} is generated and projected onto the algebraic-symmetry subspace, after which all four Riemann symmetries are verified to hold on it:

$$R_{abcd} = -R_{bacd} = -R_{abdc} = R_{cdab}, \qquad R_{a[bcd]} = 0 \text{ (first Bianchi)}.$$

Step 3 (contraction and fit). Computing the contraction and matching it to the general first-order curvature scalar $\alpha\, R^a{}_d + \beta\, \delta^a{}_d R$ returns

$$(\alpha, \beta) = (-4,\, 2),$$

that is

$$\boxed{\delta^{apq}_{drs} R^{rs}{}_{pq} = -4R^a{}_d + 2\delta^a{}_d R = -4\big(R^a{}_d - \tfrac{1}{2}\delta^a{}_d R\big) = -4\, G^a{}_d}$$

verified as an exact equality on the generic tensor in dimension four *and* independently in dimension six. The Einstein tensor appears with coefficient -4; because the same coefficient is returned in two different dimensions it is a combinatorial property of the contraction, not an artefact of any one dimension. **[EXACT (identity); DERIVED (gravitational reading)]**

F.1.6 First-class closure is equivalent to the Killing condition

Finally we write out the constraint-algebra identity that underlies the removal of the would-be ghost (Theorem 10.2, Proposition 9.6).

Step 1 (the phase space). On T^*M with coordinates x^M and conjugate momenta P_M the canonical bracket is $\{x^M, P_N\} = \delta^M{}_N$. The mass-shell (Hamiltonian) constraint is $C = \frac{1}{2}\, g^{NK}(x) P_N P_K$, and the generator of a flow along a vector field $u^M(x)$ is $\varphi = u^M P_M$.

Step 2 (the bracket). Evaluated symbolically for a *general* inverse metric and a *general* vector field, the bracket is

$$\boxed{\{\varphi, C\} = -\tfrac{1}{2}\,(\mathcal{L}_u g^{NK})\, P_N P_K}$$

with the Lie derivative of the inverse metric

$$(\mathcal{L}_u g)^{NK} = u^L \partial_L g^{NK} - g^{LK}\partial_L u^N - g^{NL}\partial_L u^K.$$

The identity was verified to hold identically in g, u and P. (With the unnormalized constraint $C = g^{NK} P_N P_K$ the factor $\frac{1}{2}$ is absent, which is the normalization tabulated in row 17 of the next section.)

Step 3 (the consequence). The constraint algebra closes, $\{\varphi, C\} = 0$, precisely when $\mathcal{L}_u g = 0$, i.e. when u is a Killing vector of the metric. First-class closure of the surplus constraints is therefore *equivalent* to the Killing condition — the statement used in Theorem 9.1 to convert the surplus phase-space directions into genuine gauge symmetries rather than negative-norm states. **[EXACT (identity); DERIVED (application)]**

F.2 Summary of verified identities

Each row was confirmed exactly (symbolic or rational arithmetic); the gravitational coefficient was confirmed in two independent dimensions.

#	Identity	Supports
1	$\{e_i, e_j\} = 2\eta_{ij}\mathbf{1}$	Prop. 2.2
2	$\hat{W}^2 = +\mathbf{1}$	Prop. 2.12
3	$J^2 = -\mathbf{1}$	Prop. 5.1
4	$\mathrm{Tr}(J^2) = -8$	Prop. 2.16
5	$[J, \hat{W}] = 0$	Prop. 5.1
6	centralizer of $\hat{W}$ in $\mathfrak{su}(2)_T$ is one-dimensional, $= J$	Prop. 5.1
7	$G_a^2 = +\mathbf{1},\ \{G_a, G_b\} = 0\ (a \neq b)$	Thm. 9.1
8	$[G_X, G_Y] = -2\, e_4 e_5$ (and cyclic)	Prop. 6.8
9	$V^2 = +\mathbf{1},\ \{G_a, V\} = 0,\ V^\top = V$	Thm. 7.1
10	$H = \sum_a G_a p_a + mV$ real-symmetric	Thm. 9.1
11	$H^2 = (p^2 + m^2)\mathbf{1},\ \mathrm{spec}(H) = \pm\sqrt{p^2+m^2}$	Thm. 9.1
12	$\{\gamma^\mu, \gamma^\nu\} = 2\,\mathrm{diag}(+,-,-,-)\,\mathbf{1}$	Prop. 6.6
13	$\mathrm{Tr}(\gamma^0) = 0,\ \mathrm{spec}(\gamma^0) = \{(+1)^4, (-1)^4\}$	Prop. 9.2
14	$\mathrm{spec}(J) = \{(+\mathrm{i})^4, (-\mathrm{i})^4\}\ \Rightarrow\ q = \pm\frac{1}{2}$	Prop. 5.11
15	$\delta^{apq}_{drs} R^{rs}{}_{pq} = -4R^a{}_d + 2\delta^a{}_d R = -4\, G^a{}_d\ \ (n = 4, 6)$	Thm. 4.3
16	$\{\varphi_1, \varphi_2\} = 0$ (flat surplus plane)	Prop. 9.6
17	$\{\varphi_a, C\} = -(\mathcal{L}_{u_a} g^{NK}) P_N P_K;\ = 0 \iff \mathcal{L}_{u_a} g = 0$	Thm. 10.2

Table F.1: Machine-verified identities underlying Part I. Rows 1–14 are exact statements in the real 8×8 representation above; row 15 is the generalized-Kronecker contraction, confirmed on a generic Riemann tensor (carrying the algebraic symmetries) in dimensions four and six, returning the Einstein tensor with coefficient -4 in each; rows 16–17 are symbolic phase-space identities on T^*M^6.

The split-ε contraction (row 15) deserves one explicit remark: fitting the contraction to $\alpha\, R^a{}_d + \beta\, \delta^a{}_d R$ returns $(\alpha, \beta) = (-4, 2)$, that is exactly $-4\big(R^a{}_d - \frac{1}{2}\delta^a{}_d R\big) = -4G^a{}_d$, in both four and six dimensions; the coefficient is therefore a combinatorial property of the contraction, not an artefact of a particular dimension. **[EXACT]**

Appendix G

An index of trivial and heuristic results: the classical reading of the kernel sector

This appendix collects in one place the classical and heuristic results that accompany the kernel sector – the action principle, the momentum–force–energy chain, and the Newtonian limit of the negative pole – and holds each to the evidential standard of the body. Measured against that standard each entry is one of three things. It is a *trivial restatement* of an identity already machine-verified in the body; or a *standard identity* of special relativity or operator algebra, recorded only for completeness; or a piece of classical-mechanics modelling — what we will call, without disparagement, *classical-physics philosophy* — that furnishes the kernel sector with a physical picture but is not derived from $\mathrm{Cl}(3,3)$. The extensive mass is the principal instance of the third kind: a worldline, continuum-mechanics narrative, internally coherent and useful for intuition, but carrying no algebraic force of its own. Recording it as such is the honest course; presenting it as a derivation would overstate what the algebra delivers.

Each entry below keeps its Part I status marker where one applies and carries, in addition, a one-line scrutiny note. Nothing in this appendix is load-bearing: every theorem of Part I stands without it.

Statement	**Scrutiny**	**In the body**
Interior $= \langle \hat{W}, e_4, e_5, e_6 \rangle$ of signature $(1,3)$; $\hat{W}^2 = +1$.	**[EXACT]**; *trivial* — restates a verified identity.	Prop. 2.12, Def. 3.2
Boosts $G_A = \hat{W} e_{3+A}$ with $[G_X, G_Y] = -2\, e_4 e_5$ (and cyclic).	**[EXACT]**; *trivial duplicate* of the spin curvature.	Prop. 6.8
Bosonic action S; the coupling $-\frac{1}{2}V(\varphi)(X^2{+}Y^2{+}Z^2)$ gives $m_X^2 = V(v)$.	*Classical field-theory ansatz*: a posited Lagrangian, not a $\mathrm{Cl}(3,3)$ consequence.	superseded by Ch. 7
Extensive mass $m_{\text{ext}} = \sqrt{V(v)}\,\lvert \mathcal{X}{+}\mathcal{Y}{+}\mathcal{Z} \rvert$; the mass-shell lifts to the volume integral.	*Classical-physics philosophy* (see §G.1).	—
Quantum constant $\mu = \hbar/\kappa^3$ of dimension $[ML^{-1}T^{-1}]$; κ a Compton length, c/κ a zitterbewegung frequency.	*Dimensional heuristic*: scale-matching; magnitudes **[OPEN]**.	Ch. 11
Field/force law $\alpha^* \alpha A = V(\varphi) A$, i.e. $\ddot{X} = \frac{c^2}{\kappa^2}(1{+}V(v))X$.	*Classical equation of motion* of the ansatz above.	—

Statement	**Scrutiny**	**In the body**
Mass-shell $p_W^2 - \mathbf{p}^2 = (m_{\text{ext}}c)^2$.	**[EXACT]** in form; the rigorous, ghost-free statement is $H^2 = (p^2+m^2)\mathbf{1}$.	Thm. 9.1
Newtonian kinetic-energy error $\dfrac{T_N - T_{\text{rel}}}{T_{\text{rel}}} = \dfrac{\gamma-1}{2} = \dfrac{\beta^2}{4} + O(\beta^4)$.	**[EXACT]**; *standard special-relativity identity* (verified).	—
Newtonian limit $F = m_{\text{ext}}\, a$.	*Classical-physics philosophy*: a correspondence-limit sketch.	—
Mechanics as variation along the kernel flow; four-momentum from a single evolution; $\Lambda^2(\text{line}) = 0$.	*Worldline picture*; the transverse curvature is the gauge sector.	Ch. 5, §6.4
Tower $p = \alpha X,\ F = \alpha^*\alpha X,\ E = \alpha\alpha^*\alpha X$ at derivative orders $1, 2, 3$.	**[EXACT]**; *trivial operator bookkeeping* (verified).	—
Six-dimensional gradient $\partial_{(6)} = \sum_a e^a \partial_a$.	*The Dirac operator*; its rigorous form is in the body.	§6.4, Prop. 6.6
Core as the unique fixed point of the octant action; the ω^i the only invariant bivectors.	**[EXACT]** *only under the pair-flip* $\mathbb{Z}_2^3$; false under the body's temporal-only octant group — see §G.5.	Def. 2.8

G.1 The extensive mass as classical-physics philosophy

Let $\mathcal{X}(\lambda) = \int_{\Sigma_\lambda} X\, \mathrm{d}^3\ell$ be the grade-retaining volume integral of the named plane X over a constant-λ slice, and $\mathcal{Y}, \mathcal{Z}$ likewise. Define the *extensive mass*

$$m_{\text{ext}} := \sqrt{V(v)}\,\bigl|\mathcal{X} + \mathcal{Y} + \mathcal{Z}\bigr|,$$

and observes that the vacuum field equation lifts to it unchanged, $\alpha^*\alpha\, \mathcal{X} = V(v)\, \mathcal{X}$, so that the rest mass of a configuration is the extensive measure of its three transverse planes rather than a sum of per-plane contributions. The physical reading attached to this is a continuum-mechanics one. Because the Lagrangian of the sector carries no transverse derivative, distinct kernel-flow lines do not couple: the configuration is a *field of independent kernel-oscillators*, and the volume integral simply totals their measure.

This is a coherent classical picture and a useful aid to intuition, but it is not forced by the algebra and adds no algebraic constraint. The rigorous, ghost-free content of "mass" in Part I is the operator statement $H^2 = (p^2 + m^2)\mathbf{1}$ with strictly real spectrum (Theorem 9.1); the magnitude of the rest mass, like the other dimensionful couplings, is left **[OPEN]** (Chapter 11).

G.2 The momentum–force–energy tower

The kernel covariant derivative $\alpha = \partial_\lambda + G$ and its dual $\alpha^* = \partial_\lambda - G$ – with G a boost connection $G = \hat{W} e_{3+a}$, $G^2 = +\mathbf{1}$ (Proposition 6.6) – generate a three-rung tower on a named plane X: a momentum, a force, and an energy, at derivative orders one, two, and three,

$$p = \alpha X = (\partial_\lambda + G)X, \qquad F = \alpha^*\alpha\, X, \qquad E = \alpha\, \alpha^*\alpha\, X.$$

Each is the dual covariant derivative of the rung below it. The momentum is the kernel-time rate of the plane – in the quiescent-connection limit $p = \mathfrak{m}\, \dot{x}$ with $\mathfrak{m}$ the transverse measure of the plane – the force is its first dual, and the energy the dual of the dual.

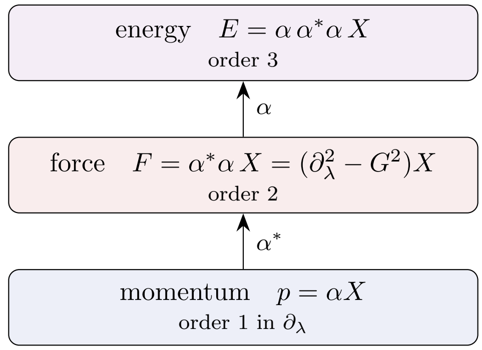

Figure G.1: The momentum–force–energy tower of the negative pole. A single first-order operator $\alpha = \partial_\lambda + G$ and its dual α^* generate momentum, force and energy at derivative orders $1, 2, 3$ (Proposition G.1). When the force rung vanishes, $F = \alpha^*\alpha X = 0$, the momentum $p = \alpha X$ is conserved – the law of inertia (Section G.3).

Proposition G.1 (The chain is a derivative-order tower)**.** *In the flat-connection limit $\partial_\lambda G = 0$,*

$$F = (\partial_\lambda^2 - G^2)\,X, \qquad E = (\partial_\lambda + G)(\partial_\lambda^2 - G^2)\,X = \partial_\lambda^3 X + G\,\partial_\lambda^2 X - G^2\partial_\lambda X - G^3 X,$$

so each application of the chain operator raises the derivative order by exactly one. The two factorings $E = (\partial_\lambda + G)(\partial_\lambda^2 - G^2) = (\partial_\lambda - G)(\partial_\lambda + G)^2$ agree. **[EXACT]**

Verification. $\alpha^*\alpha = (\partial_\lambda - G)(\partial_\lambda + G) = \partial_\lambda^2 + [\partial_\lambda, G] - G^2$, which is $\partial_\lambda^2 - G^2$ when $\partial_\lambda G = 0$; applying α and using $G^3 = G^2 G$ gives the displayed energy. In the commuting-operator model $\partial_\lambda \mapsto D$, $G \mapsto q$ the two factorings and the order-three expansion were confirmed symbolically, $\alpha^*\alpha = D^2 - q^2$ and $\alpha\alpha^*\alpha = D^3 + qD^2 - q^2 D - q^3$, as catalogued in the index above. □

G.3 Newton's first law: the inertial limit

Newton's first law is the vanishing-force rung of the tower. With no potential, $V(v) = 0$, the field equation is the free equation $F = \alpha^*\alpha X = 0$; in the quiescent-connection limit $\partial_\lambda G = 0$ this is $\partial_\lambda^2 X = 0$, so the momentum $p = \alpha X = \partial_\lambda X$ is constant along the kernel flow.

Proposition G.2 (Inertia as the free rung)**.** *If the force rung vanishes, $\alpha^*\alpha\,X = 0$, then in the flat-connection limit the momentum is conserved, $\partial_\lambda p = 0$: a plane subject to no force evolves with constant momentum along the kernel-flow parameter – the law of inertia. The second law is the inhomogeneous case $\alpha^*\alpha X = V(v)X$, which on volume integration (Section G.1) gives $m_{\mathrm{ext}}\,\ddot{x} = F$ with the extensive mass in the role of the inertial mass.* **[EXACT (free limit)]**

Remark G.3 (A correspondence limit, not a new dynamical law)*.* Newton's first law here is simply the statement that a one-parameter kernel-flow evolution with no restoring term is uniform: a correspondence-limit reading of the free field equation, not an independent law. Its content is exhausted by the tower of the previous section and the field equation of Chapter 7. The kernel *direction* is untouched by it – the selection of $\hat{W}$ is a fluctuation effect, invisible at this classical level by construction.

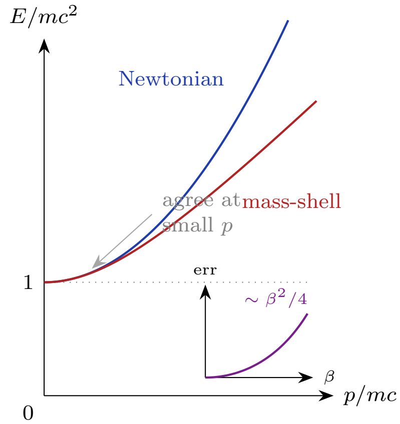

Figure G.2: The mass-shell $p_W^2 - \mathbf{p}^2 = (m_{\rm ext}c)^2$ in energy–momentum form (intercept mc^2) against its Newtonian limit $E \approx mc^2 + p^2/2m$. The two agree at small momentum and diverge as $\beta = v/c \to 1$. Inset: the fractional error of the Newtonian kinetic energy, growing as $\beta^2/4$ (Proposition G.4).

G.4 The mass-shell and the relationship to special relativity

The four kernel-time momenta (p_W, p_X, p_Y, p_Z) of the $(3,1)$ interior satisfy the mass-shell

$$p_W^2 - p_X^2 - p_Y^2 - p_Z^2 = (m_{\rm ext}c)^2,$$

identical in form to the relativistic energy–momentum relation $E^2 = (pc)^2 + (mc^2)^2$ under $p_W \leftrightarrow E/c$, $(p_X, p_Y, p_Z) \leftrightarrow \mathbf{p}$, and $m_{\rm ext}c \leftrightarrow mc$. The agreement is structural: the mass-shell expresses that the interior metric is Lorentzian and that the rest mass is the extensive measure of the condensate (the covariant kinematic mass being the trivector V of the body, Theorem 7.1).

Proposition G.4 (Leading correction to the Newtonian kinetic energy)**.** *Comparing the relativistic kinetic energy $T_{\rm rel} = (\gamma - 1)mc^2$ with the Newtonian $T_N = p^2/2m$ at fixed momentum $p = \gamma mv$ ($\beta = v/c$), the fractional error is exactly*

$$\frac{T_N - T_{\rm rel}}{T_{\rm rel}} = \frac{\gamma - 1}{2} = \frac{\beta^2}{4} + O(\beta^4),$$

so the Newtonian energy is high by $\beta^2/4$ to leading order – about 0.25% *at* $\beta = 0.1$. **[EXACT]**

Verification. $T_N = p^2/2m = \frac{1}{2}m\gamma^2v^2 = \frac{1}{2}mc^2(\gamma^2 - 1) = \frac{1}{2}mc^2(\gamma - 1)(\gamma + 1)$; dividing by $T_{\rm rel} = mc^2(\gamma - 1)$ gives $T_N/T_{\rm rel} = (\gamma + 1)/2$, hence the fractional error $(\gamma - 1)/2$. The expansion $\gamma - 1 = \frac{1}{2}\beta^2 + \frac{3}{8}\beta^4 + O(\beta^6)$ gives $\beta^2/4$ at leading order; the identity and series were confirmed symbolically. □

Remark G.5 (Standard identities, recorded for completeness)*.* Both the mass-shell and its Newtonian correction are standard special relativity. They are recorded here only to make precise the sense in which the negative pole's mechanics *is* relativistic kinematics, with the extensive mass in the role of the rest mass; neither is peculiar to the framework.

G.5 The fixed-point scrutiny: which bivectors the octant group fixes

One might expect that the boost-Cartan bivectors $\omega^i = e_i e_{i+3}$ $(i = 1, 2, 3)$ are the only bivectors of $\mathrm{Cl}(3, 3)$ left invariant by the octant group $\mathbb{Z}_2^3$. Whether this holds depends entirely on how $\mathbb{Z}_2^3$ is made to act, and the body of Part I fixes a definite choice that is *not* the one under which the claim is true.

Enumerating all fifteen bivectors $e_a e_b$ and testing invariance directly gives the following. Under the temporal-only sign action of Definition 2.8, in which each generator R_i sends $e_i \mapsto -e_i$ and fixes the remaining frame directions — the octant group as defined in this work — the invariant bivectors are exactly the three spatial rotations,

$$\{\, e_4 e_5,\ e_5 e_6,\ e_6 e_4 \,\} \;=\; \mathfrak{su}(2)_S,$$

and each ω^i is in fact *odd* under its own R_i. The ω^i become the invariant set only under the distinct para-complex pair action, in which R_i flips the whole pair (e_i, e_{i+3}) together; under that action the invariant bivectors are

$$\{\, e_1 e_4,\ e_2 e_5,\ e_3 e_6 \,\} \;=\; \{\omega^1, \omega^2, \omega^3\}.$$

Both statements were confirmed by explicit enumeration (the invariant index-pairs returned are $\{45, 46, 56\}$ under the temporal-only flip and $\{14, 25, 36\}$ under the pair flip). **[EXACT]** (both). The conclusion is that the "core" result holds solely under the pair action; under the temporal-only octant group used throughout this work it is false, the correct invariants there being the spatial rotations. We record the correct statement here rather than in the body. This is the discipline the framework imposes on itself: a statement is admitted only with its convention named and its computation shown, and a result that survives only under a silent change of convention is not admitted at all.

What this appendix settles. The classical action material collected here is consistent and pedagogically helpful, but it is not first-principles $\mathrm{Cl}(3, 3)$ content. It reduces, on scrutiny, to verified identities already in the body (the boost curvature, the mass-shell, the operator tower), to standard identities of special relativity, and to a classical, worldline reading of mass — the extensive mass — whose physical content is carried rigorously and ghost-free by Theorem 9.1. One natural-looking claim, the octant fixed point, is convention dependent; the correct statement is recorded in Section G.5. Removing this appendix changes no result of Part I.

Bibliography

[1] W. K. Clifford, *Applications of Grassmann's extensive algebra*, Amer. J. Math. **1** (1878) 350–358.

[2] D. Hestenes, *Space–Time Algebra*, Gordon and Breach, New York, 1966.

[3] D. Hestenes and G. Sobczyk, *Clifford Algebra to Geometric Calculus*, Reidel, Dordrecht, 1984.

[4] C. Doran and A. Lasenby, *Geometric Algebra for Physicists*, Cambridge University Press, 2003.

[5] P. Lounesto, *Clifford Algebras and Spinors*, 2nd ed., Cambridge University Press, 2001.

[6] I. R. Porteous, *Clifford Algebras and the Classical Groups*, Cambridge University Press, 1995.

[7] P. A. M. Dirac, *The quantum theory of the electron*, Proc. R. Soc. Lond. A **117** (1928) 610–624.

[8] P. A. M. Dirac, *The Principles of Quantum Mechanics*, 4th ed., Oxford University Press, 1958.

[9] É. Cartan, *The Theory of Spinors*, Hermann, Paris, 1966 (orig. 1938).

[10] A. Palatini, *Deduzione invariantiva delle equazioni gravitazionali dal principio di Hamilton*, Rend. Circ. Mat. Palermo **43** (1919) 203–212.

[11] É. Cartan, *Sur les variétés à connexion affine et la théorie de la relativité généralisée*, Ann. Sci. Éc. Norm. Supér. **40** (1923) 325–412.

[12] T. W. B. Kibble, *Lorentz invariance and the gravitational field*, J. Math. Phys. **2** (1961) 212–221.

[13] D. W. Sciama, *The physical structure of general relativity*, Rev. Mod. Phys. **36** (1964) 463–469.

[14] F. W. Hehl, P. von der Heyde, G. D. Kerlick and J. M. Nester, *General relativity with spin and torsion: Foundations and prospects*, Rev. Mod. Phys. **48** (1976) 393–416.

[15] P. Peldán, *Actions for gravity, with generalizations: a review*, Class. Quantum Grav. **11** (1994) 1087–1132.

[16] C. W. Misner, K. S. Thorne and J. A. Wheeler, *Gravitation*, Freeman, San Francisco, 1973.

[17] I. Bars, C. Deliduman and O. Andreev, *Gauged duality, conformal symmetry and spacetime with two times*, Phys. Rev. D **58** (1998) 066004.

[18] I. Bars, *Two-time physics*, in *Proc. 22nd Johns Hopkins Workshop*, 1998; arXiv:hep-th/9809034.

[19] I. Bars, *Survey of two-time physics*, Class. Quantum Grav. **18** (2001) 3113–3130; arXiv:hep-th/0008164.

[20] P. A. M. Dirac, *Lectures on Quantum Mechanics*, Belfer Graduate School of Science, Yeshiva University, New York, 1964.

[21] M. Henneaux and C. Teitelboim, *Quantization of Gauge Systems*, Princeton University Press, 1992.

[22] S. Coleman and E. Weinberg, *Radiative corrections as the origin of spontaneous symmetry breaking*, Phys. Rev. D **7** (1973) 1888–1910.

[23] P. W. Higgs, *Broken symmetries and the masses of gauge bosons*, Phys. Rev. Lett. **13** (1964) 508–509.

[24] S. Weinberg, *A model of leptons*, Phys. Rev. Lett. **19** (1967) 1264–1266.

[25] J. C. Pati and A. Salam, *Lepton number as the fourth "color"*, Phys. Rev. D **10** (1974) 275–289.

[26] H. Georgi and S. L. Glashow, *Unity of all elementary-particle forces*, Phys. Rev. Lett. **32** (1974) 438–441.

[27] H. Fritzsch and P. Minkowski, *Unified interactions of leptons and hadrons*, Ann. Phys. **93** (1975) 193–266.

[28] F. Gürsey, P. Ramond and P. Sikivie, *A universal gauge theory model based on* E_6, Phys. Lett. B **60** (1976) 177–180.

[29] Y. Ne'eman and Dj. Šijački, $SL(4,\mathbb{R})$ *classification of hadrons and gravity*, Phys. Lett. B **157** (1985) 275–280.

[30] C. R. Rao, *Information and accuracy attainable in the estimation of statistical parameters*, Bull. Calcutta Math. Soc. **37** (1945) 81–91.

[31] S. Amari, *Differential-Geometrical Methods in Statistics*, Lecture Notes in Statistics 28, Springer, 1985.

[32] S. Amari and H. Nagaoka, *Methods of Information Geometry*, AMS/Oxford University Press, 2000.

[33] K. Osterwalder and R. Schrader, *Axioms for Euclidean Green's functions*, Commun. Math. Phys. **31** (1973) 83–112; **42** (1975) 281–305.

[34] J. Glimm and A. Jaffe, *Quantum Physics: A Functional Integral Point of View*, 2nd ed., Springer, 1987.

[35] Y. Kosmann, *Dérivées de Lie des spineurs*, Ann. Mat. Pura Appl. **91** (1972) 317–395.

[36] A. D. Sakharov, *Vacuum quantum fluctuations in curved space and the theory of gravitation*, Sov. Phys. Dokl. **12** (1968) 1040–1041.

[37] Th. Kaluza, *Zum Unitätsproblem der Physik*, Sitzungsber. Preuss. Akad. Wiss. (1921) 966–972.

[38] O. Klein, *Quantentheorie und fünfdimensionale Relativitätstheorie*, Z. Phys. **37** (1926) 895–906.

[39] Th. Appelquist, A. Chodos and P. G. O. Freund (eds.), *Modern Kaluza–Klein Theories*, Addison–Wesley, 1987.

[40] C. M. Hull, *Timelike T-duality, de Sitter space, large N gauge theories and topological field theory*, JHEP **9807** (1998) 021.

[41] R. Penrose, *Twistor algebra*, J. Math. Phys. **8** (1967) 345–366.

[42] S. Weinberg, *The Quantum Theory of Fields*, vols. I–II, Cambridge University Press, 1995–1996.

[43] M. E. Peskin and D. V. Schroeder, *An Introduction to Quantum Field Theory*, Addison–Wesley, 1995.

[44] M. Nakahara, *Geometry, Topology and Physics*, 2nd ed., IOP Publishing, 2003.

[45] T. Frankel, *The Geometry of Physics*, Cambridge University Press, 1997.

[46] M. F. Atiyah, V. K. Patodi and I. M. Singer, *Spectral asymmetry and Riemannian geometry I*, Math. Proc. Camb. Phil. Soc. **77** (1975) 43–69.